



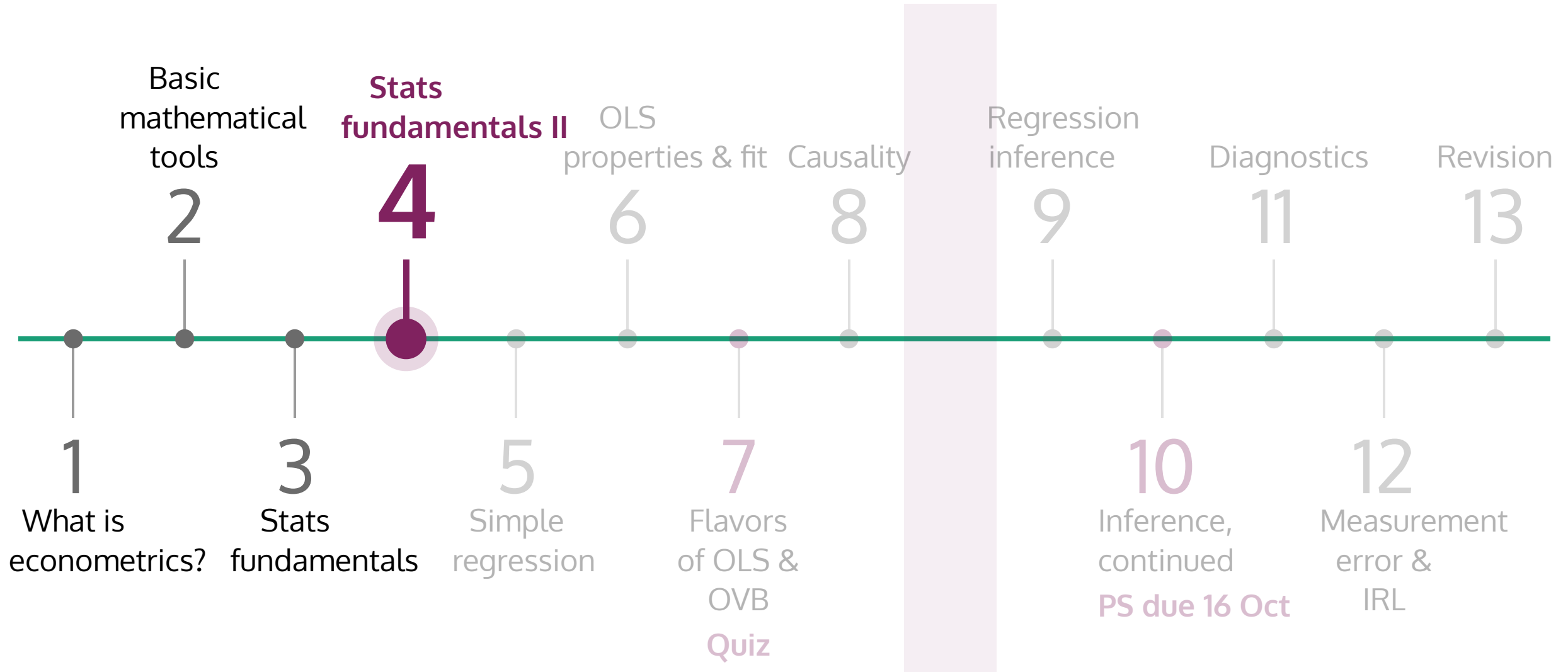
ECON 2041

# Introductory Econometrics

Great expectations



# Plan for the semester



# Plan for today

- Your grade forecasts, revealed
- Income and life expectancy in the US
- Discrete or categorical?
- The correlation gallery
- Where a covariance comes from

What you expect of yourself

# What I asked you

## Your own semester

- Expected final mark (0-100)
- Plans:
  - attend live?
  - study hours?
  - try tutorials before class?
- The mark if you did "everything"

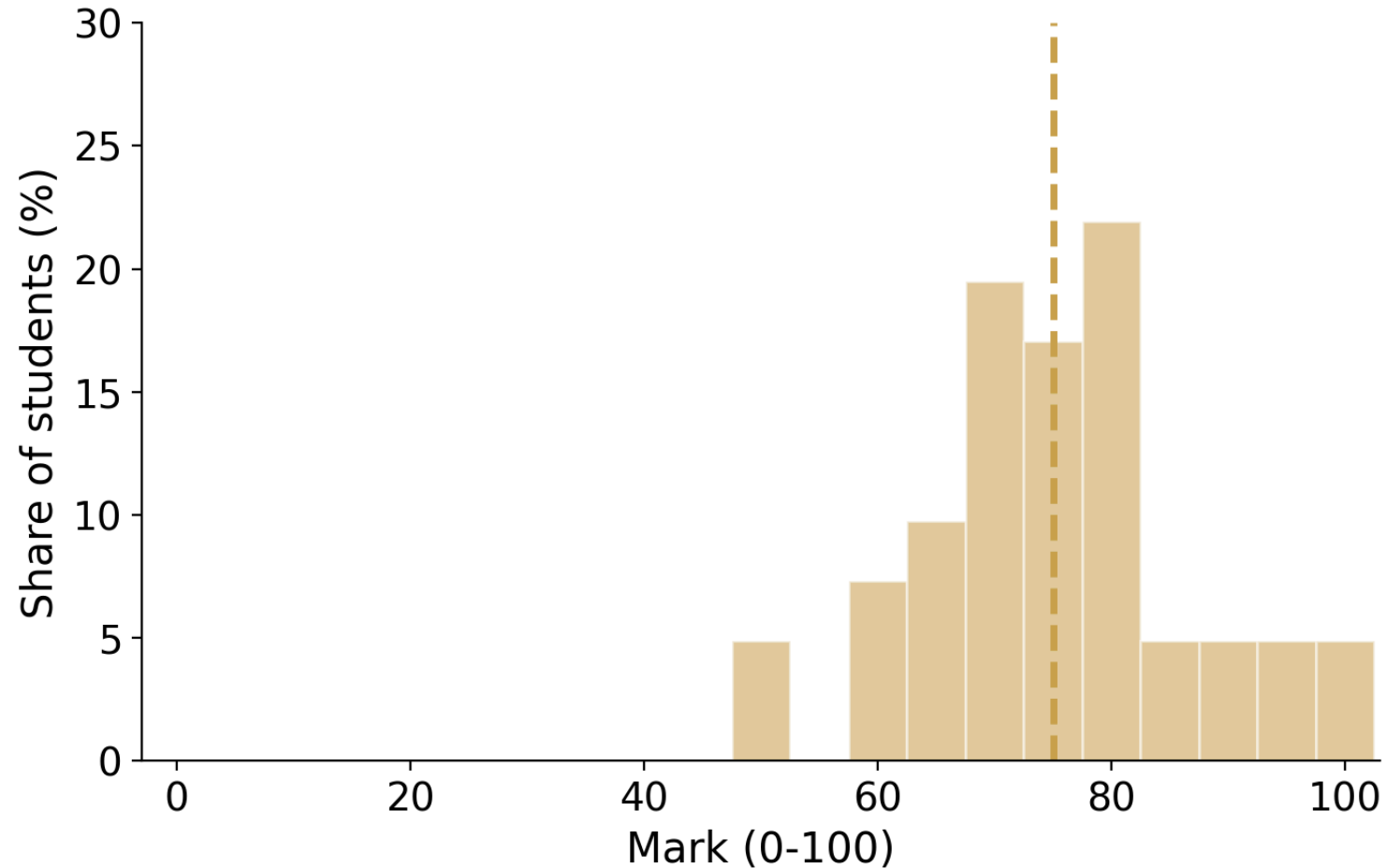
## Three hypothetical students

- Attends live lecture:
  - rarely
  - some weeks
  - most weeks
- Study hours:
  - under 2
  - 2 to 5
  - 5+
- Try tutorials first: no / yes

For each: - a mark - % chance they pass, - % chance they get a D

# Nobody expects to fail; the median is 75

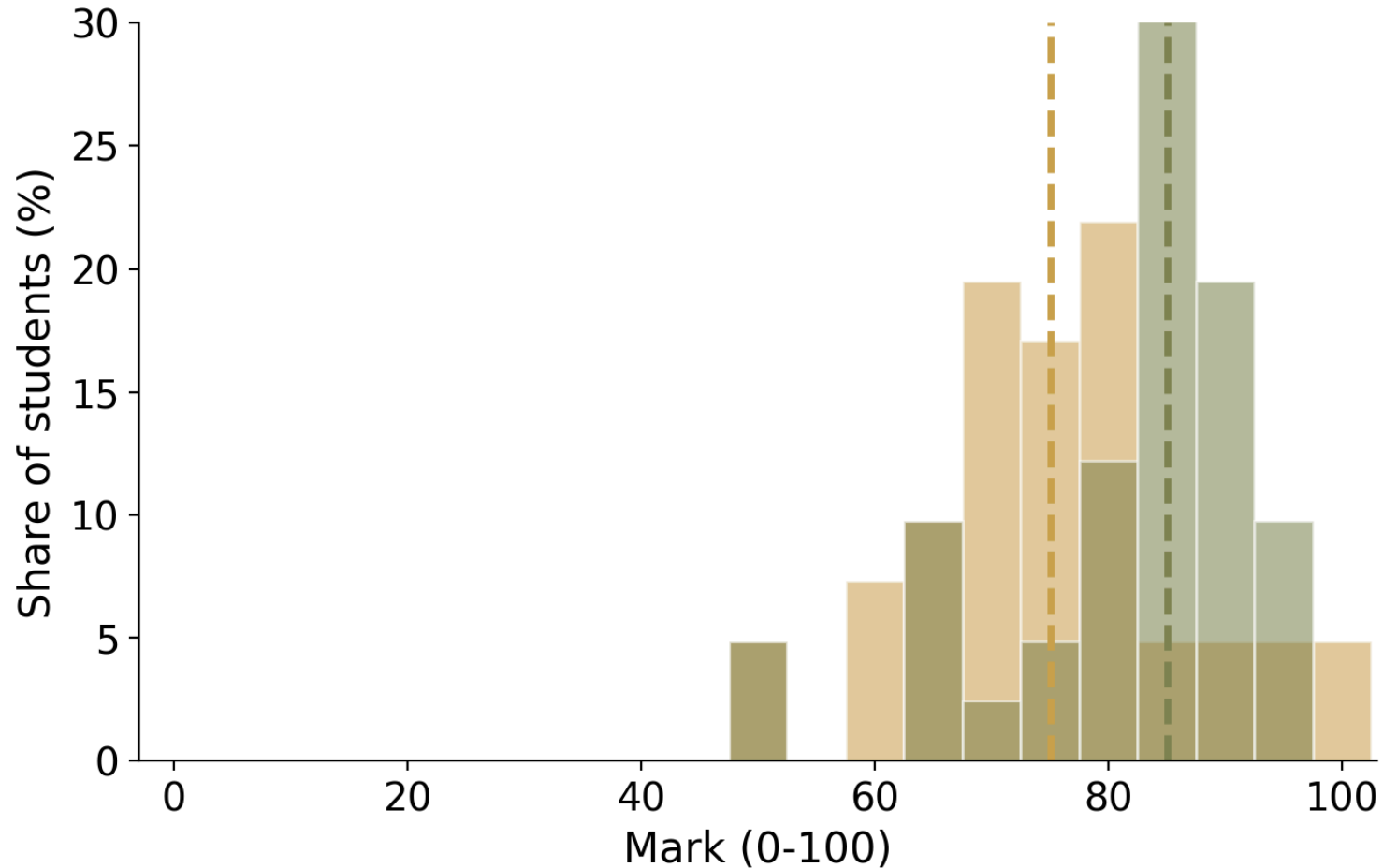
Own forecast, 2026 ( $N=41$ ), median 75



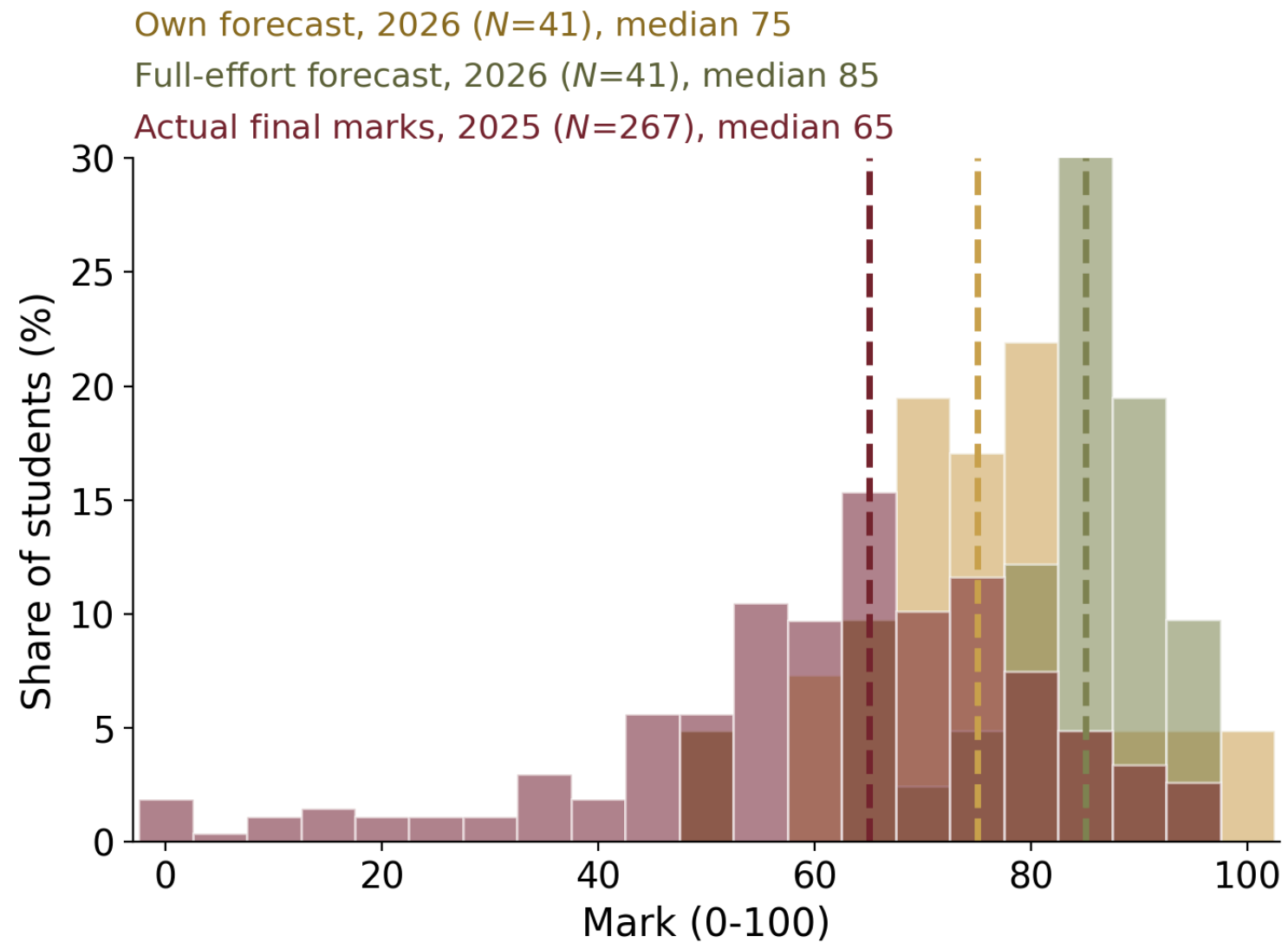
# Full effort: you expect it to add about 7 marks

Own forecast, 2026 ( $N=41$ ), median 75

Full-effort forecast, 2026 ( $N=41$ ), median 85

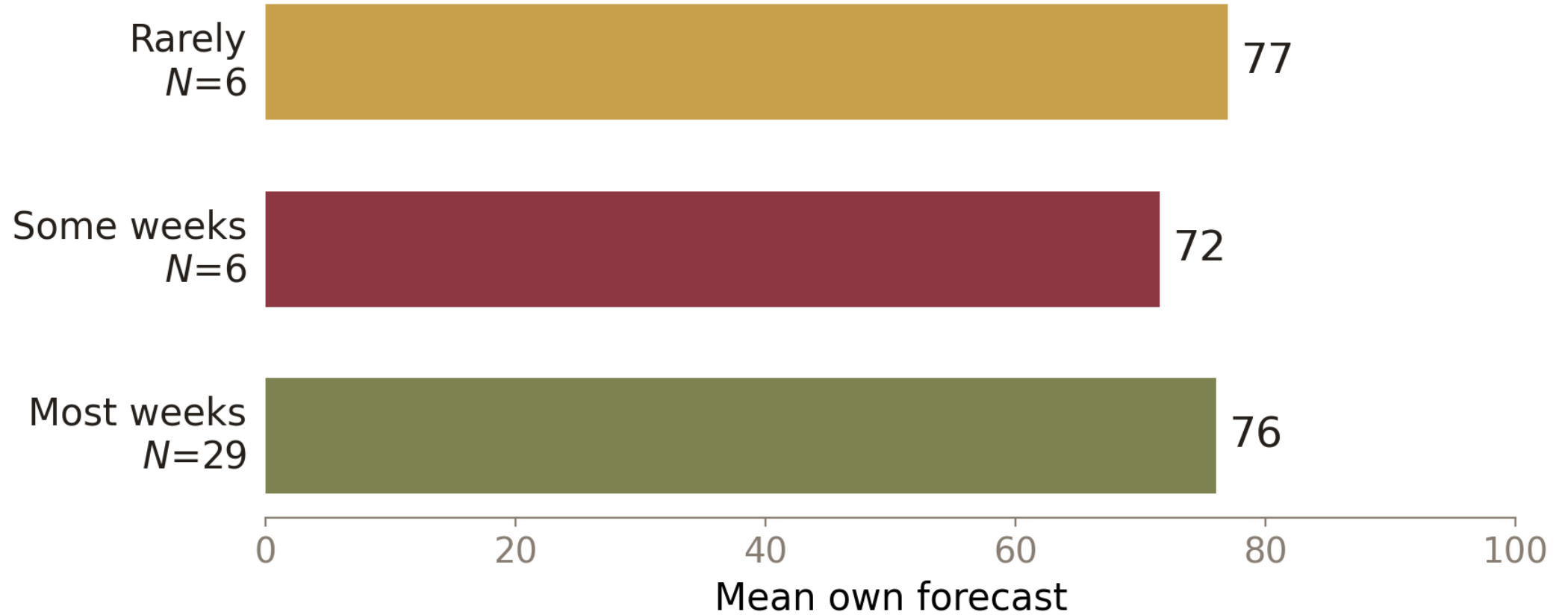


# Last year's marks: median 65, one in five failed

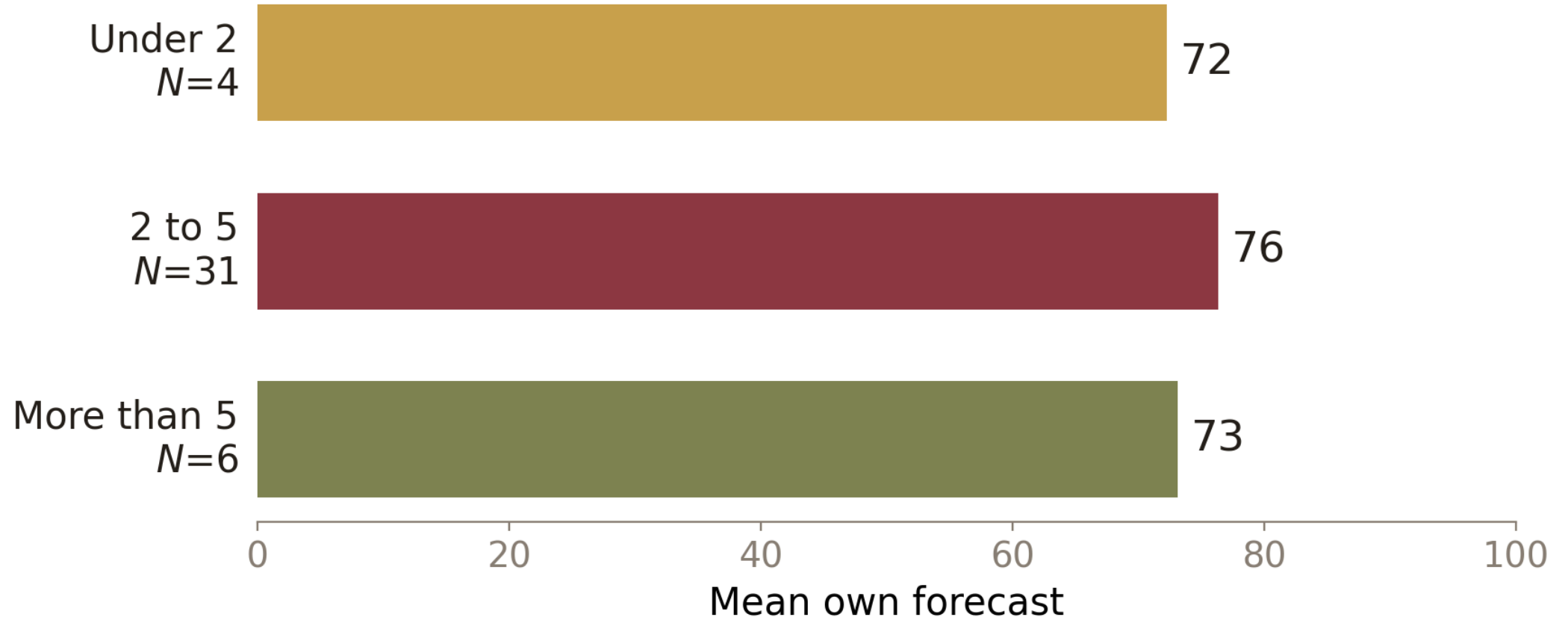




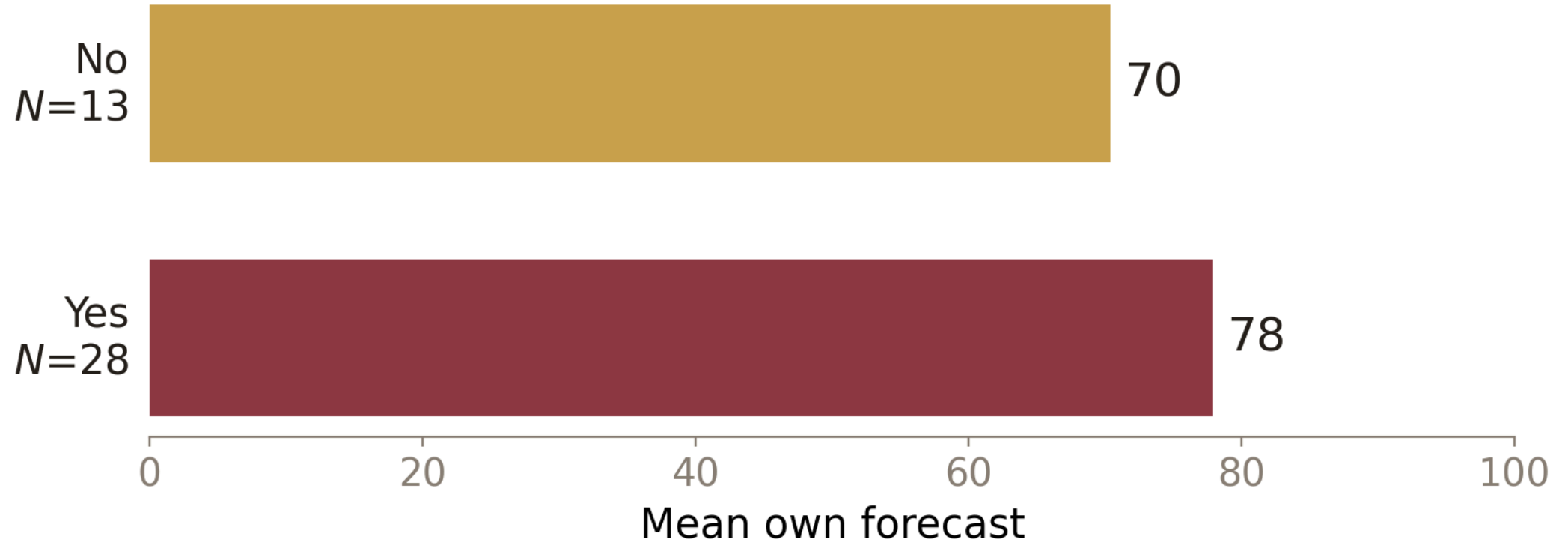
# Attendance plans: flat, on 6 and 6 people



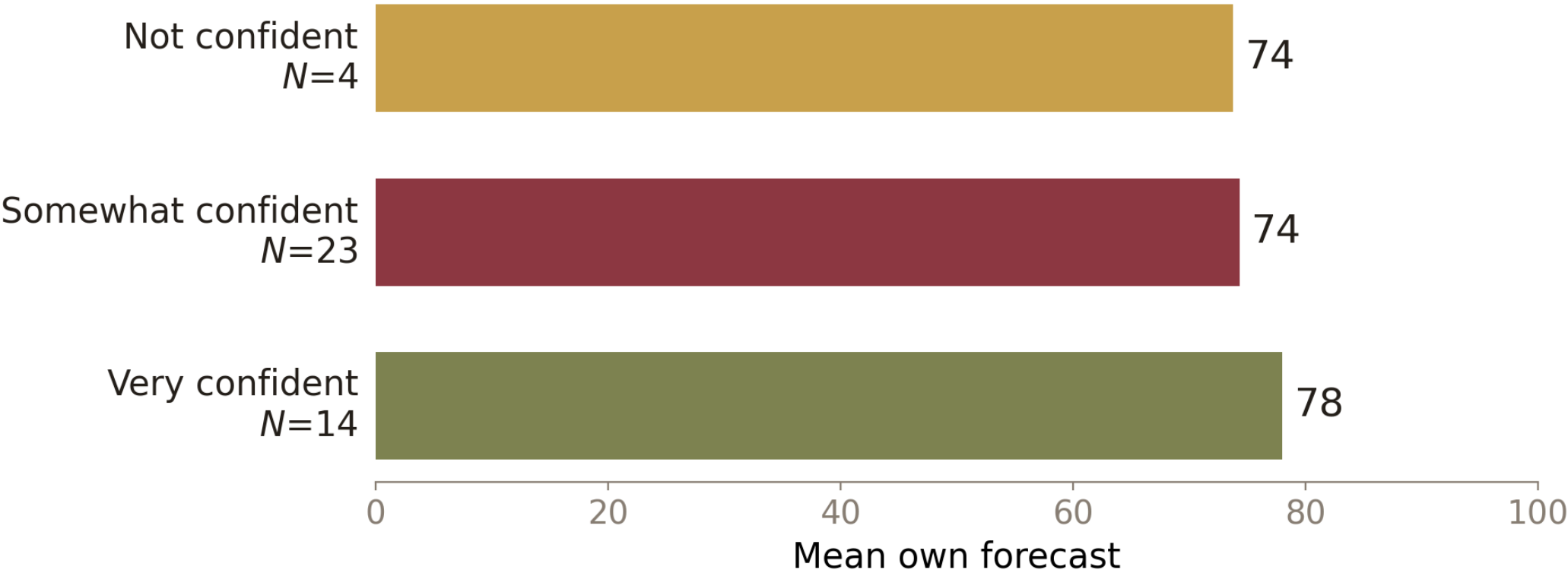
# Study-hour plans: 72, 76, 73



# Tutorials first: 70 vs 78, on 13 and 28 people

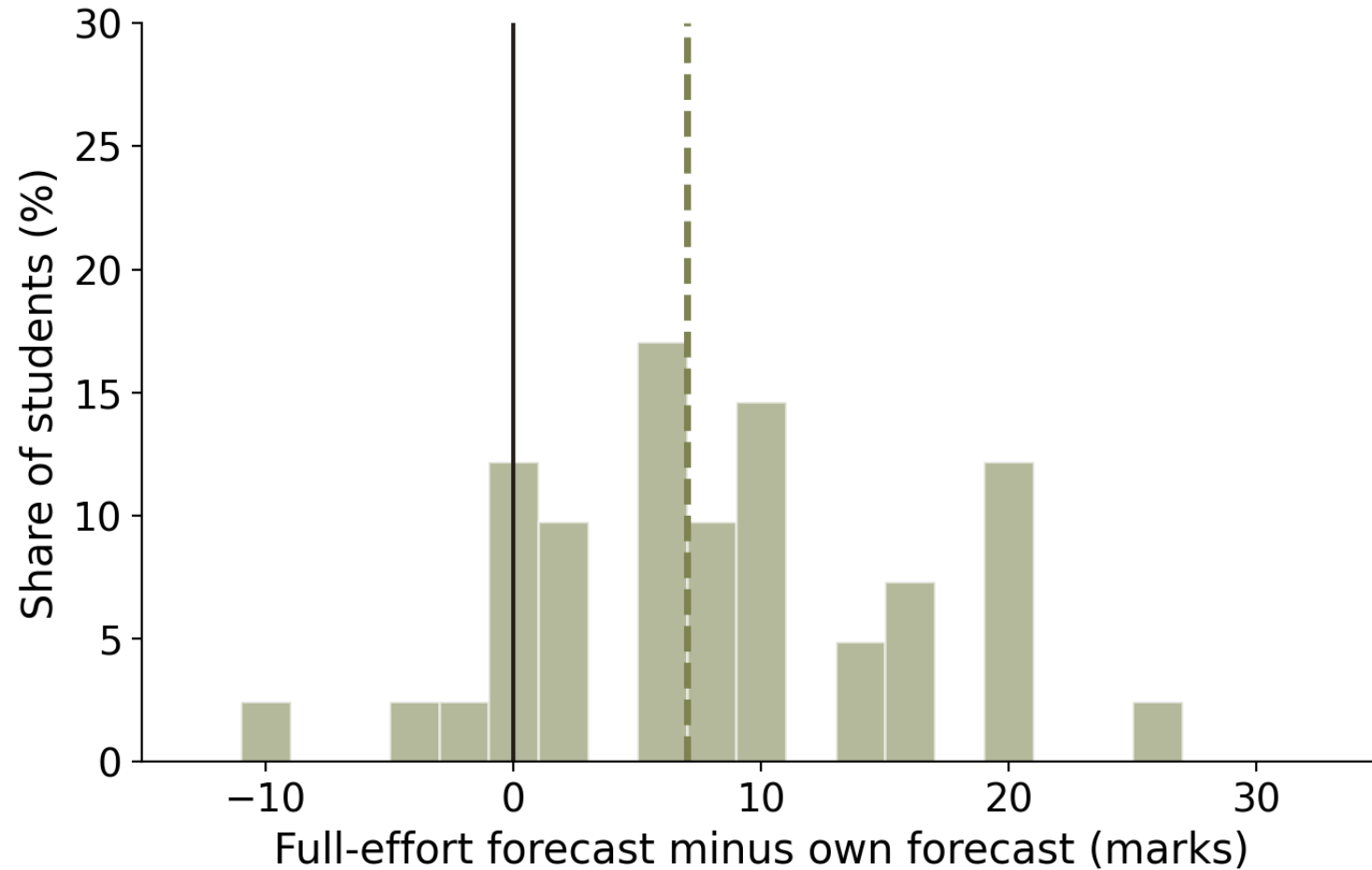


# Confidence in the plan barely matters



# Effort, student by student: median gap 7

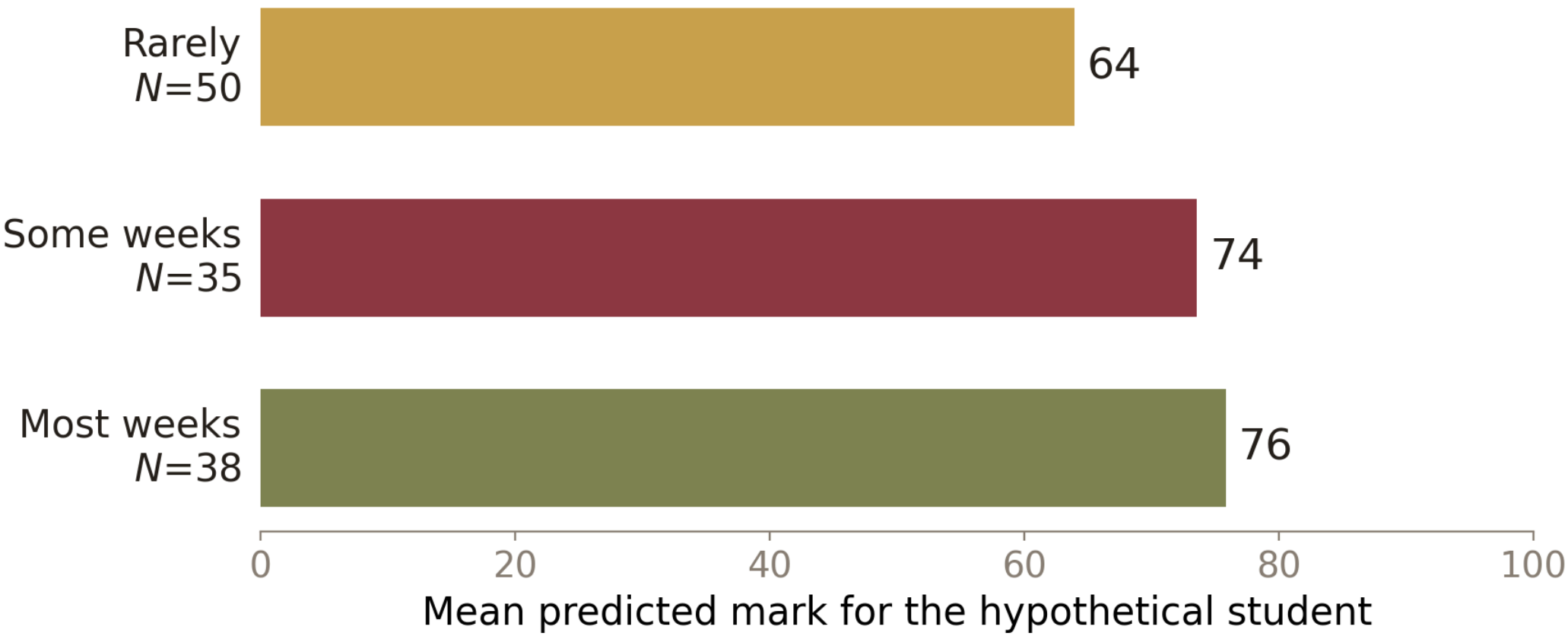
Full-effort minus own forecast, 2026 (N=41), median 7



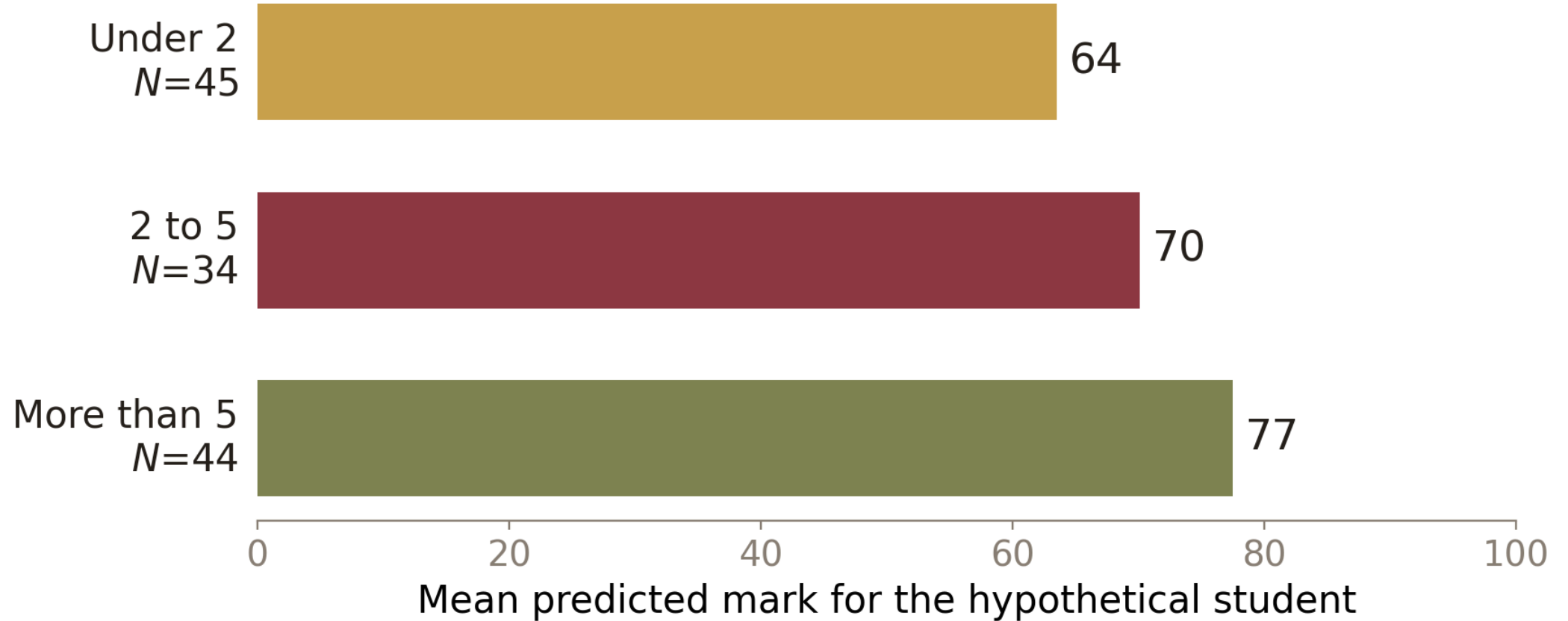


What you expect of others

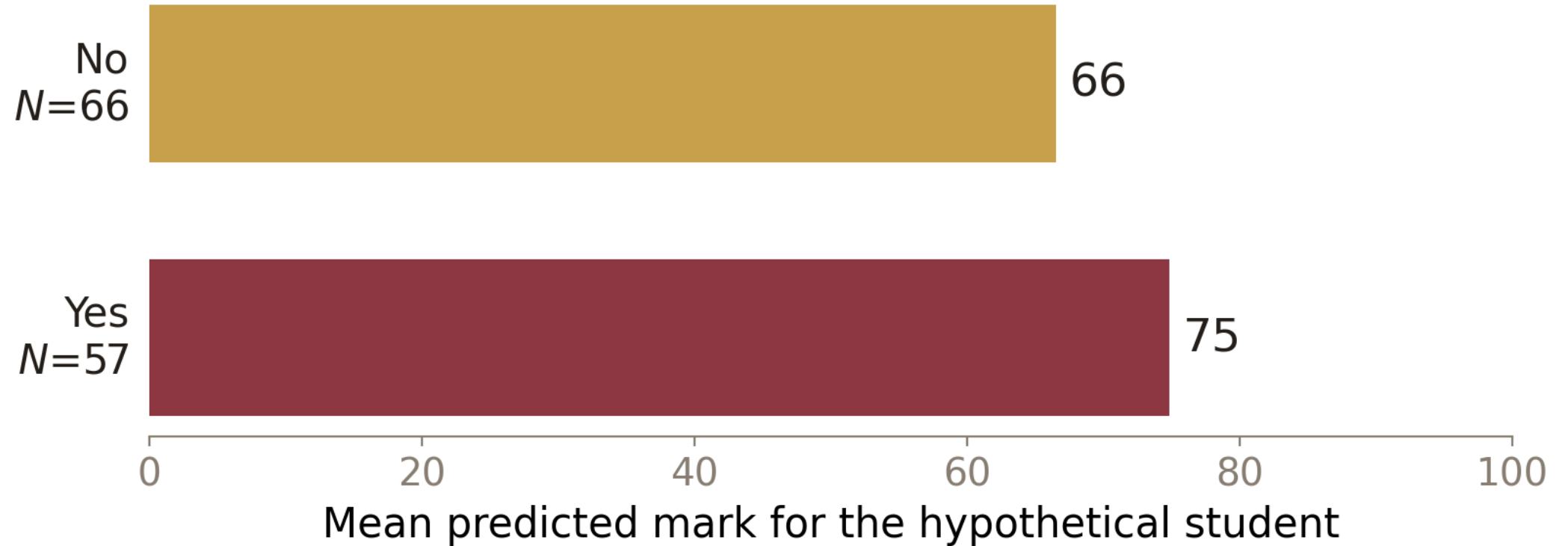
# By assigned attendance: 64, 74, 76



# Study hours: 64, 70, 78



# Tutorials before class: 67 against 75



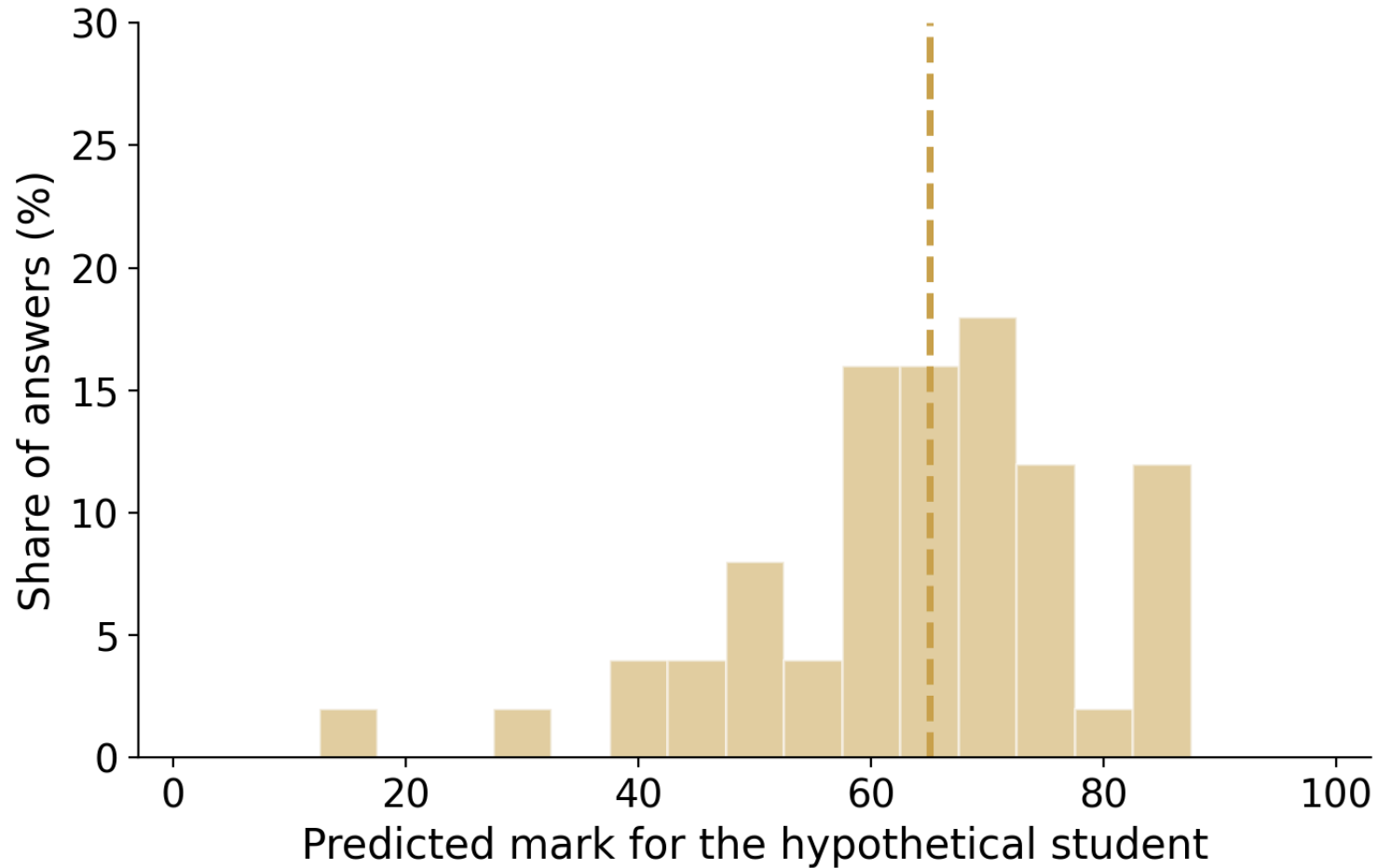
# Independence by design

Share of hypothetical students at each study-hours level, within each attendance level

| Assigned attendance | Assigned study hours per week |        |             | <i>N</i> |
|---------------------|-------------------------------|--------|-------------|----------|
|                     | Under 2                       | 2 to 5 | More than 5 |          |
| Rarely              | 44%                           | 30%    | 26%         | 50       |
| Some weeks          | 20%                           | 26%    | 54%         | 35       |
| Most weeks          | 42%                           | 26%    | 32%         | 38       |
| All vignettes       | 37%                           | 28%    | 36%         | 123      |

# Predicted marks, given "rarely"

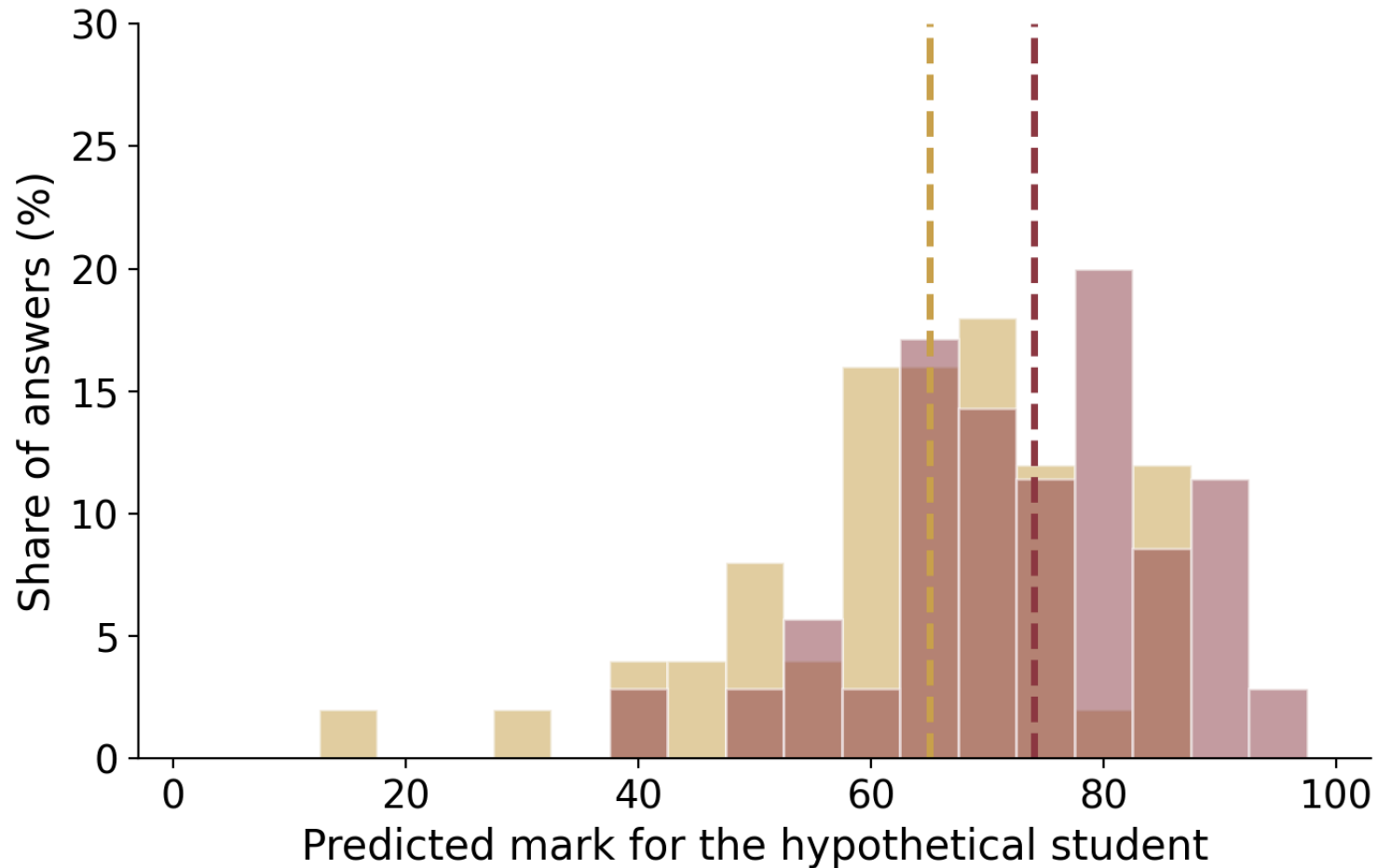
Assigned rarely ( $N=50$ ), median 65



# Predicted marks, given "some weeks"

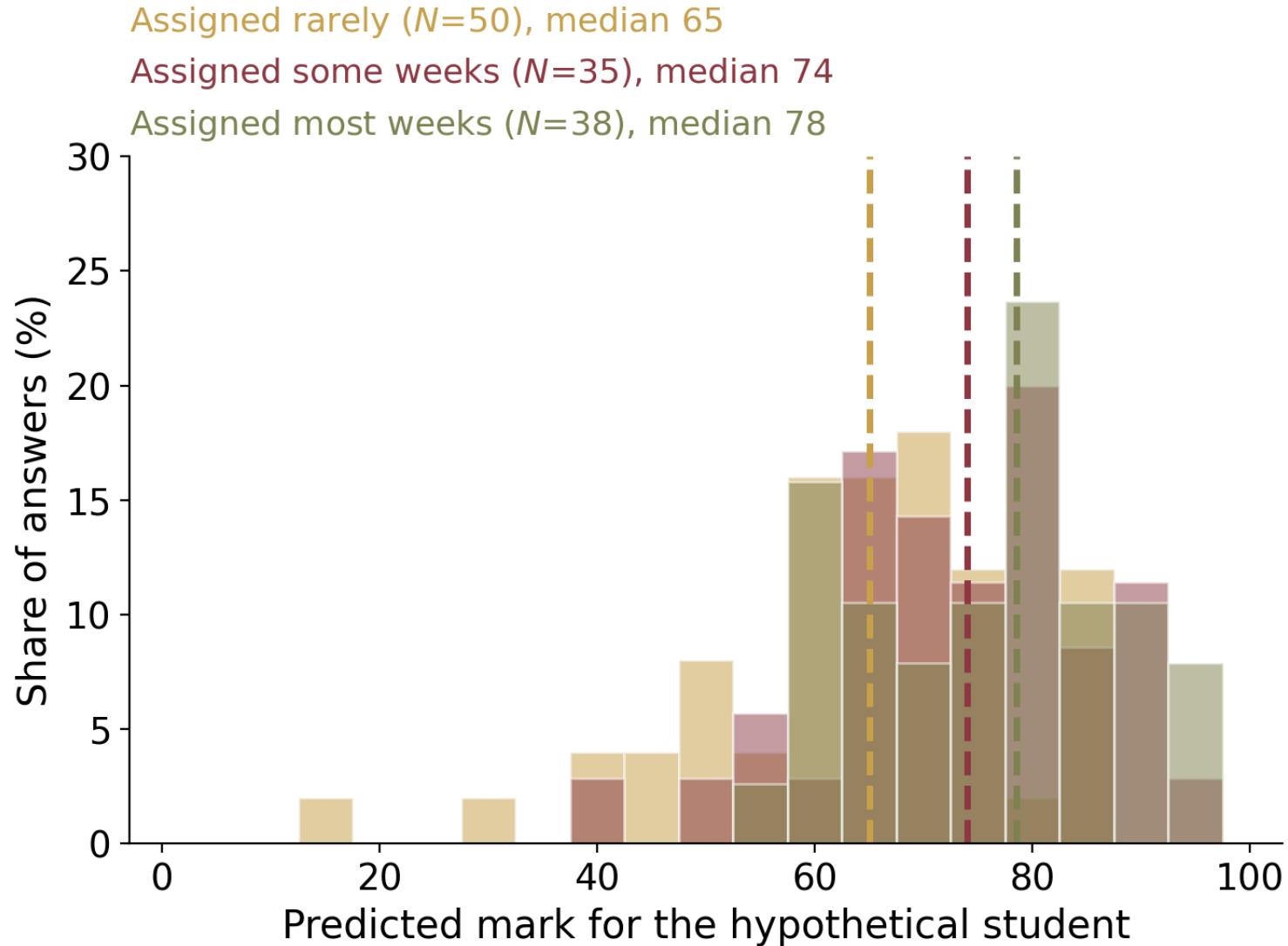
Assigned rarely ( $N=50$ ), median 65

Assigned some weeks ( $N=35$ ), median 74

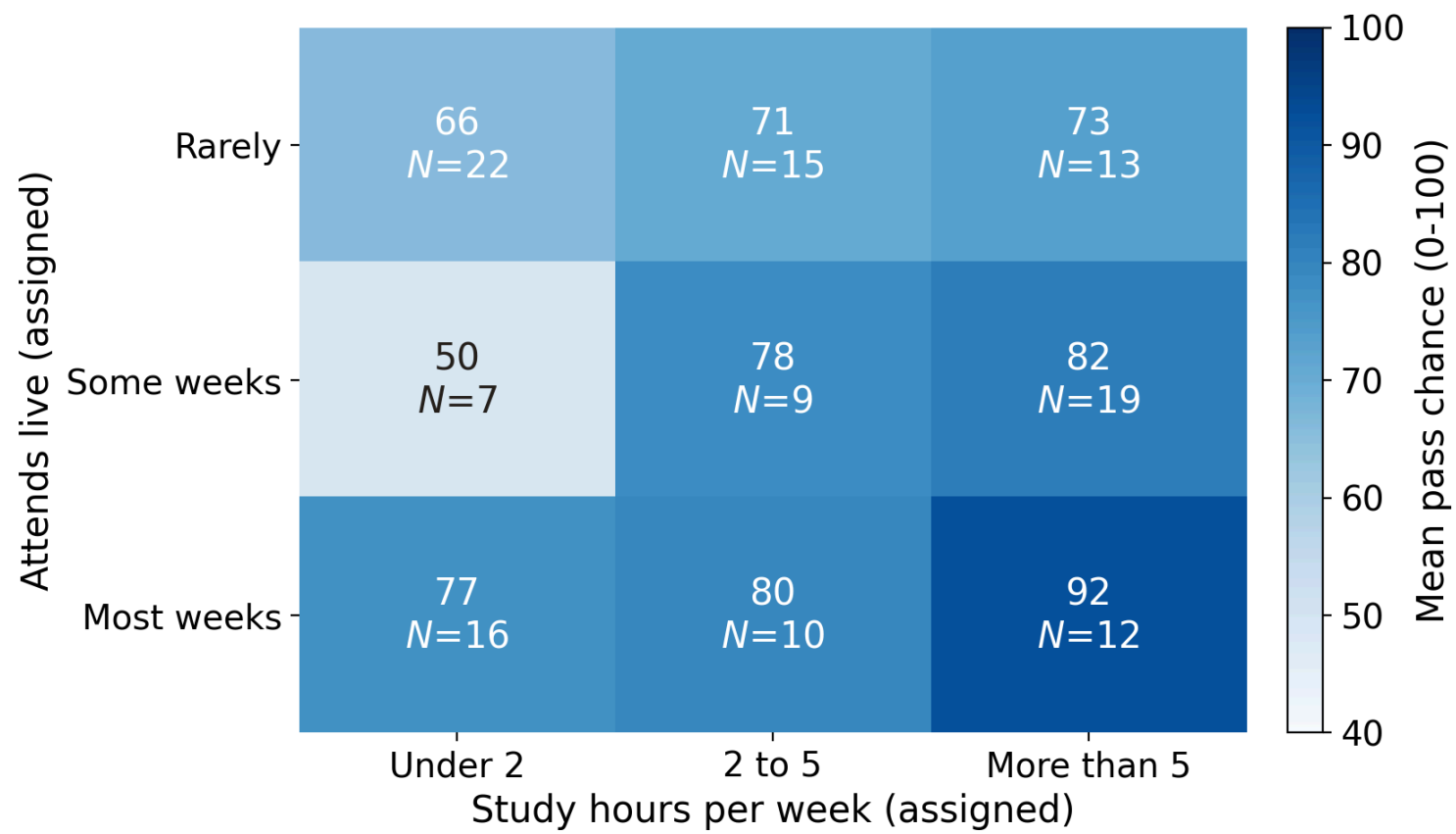




# Marks are not independent of attendance



# Pass chance, given attendance and hours



# A distinction is a pass; 10 answers disagreed

For each hypothetical student, two chances:

- $P(\text{mark} \geq 50)$ , the chance of a pass
- $P(\text{mark} \geq 75)$ , the chance of a distinction

Every mark of 75 or more is also a mark of 50 or more

$\{\text{mark} \geq 75\} \subset \{\text{mark} \geq 50\}$ , so  $P(\text{mark} \geq 75) \leq P(\text{mark} \geq 50)$

10 of 120 answers had the distinction chance above the pass chance

Income and life expectancy

# Exploratory data analysis: why?

We always start by looking at the data

- Check your understanding of the variables
- Guide subsequent analysis
- Give context to results
- Generate additional questions
- Offer simple, but possibly important answers!

# Descriptives can be incredibly powerful

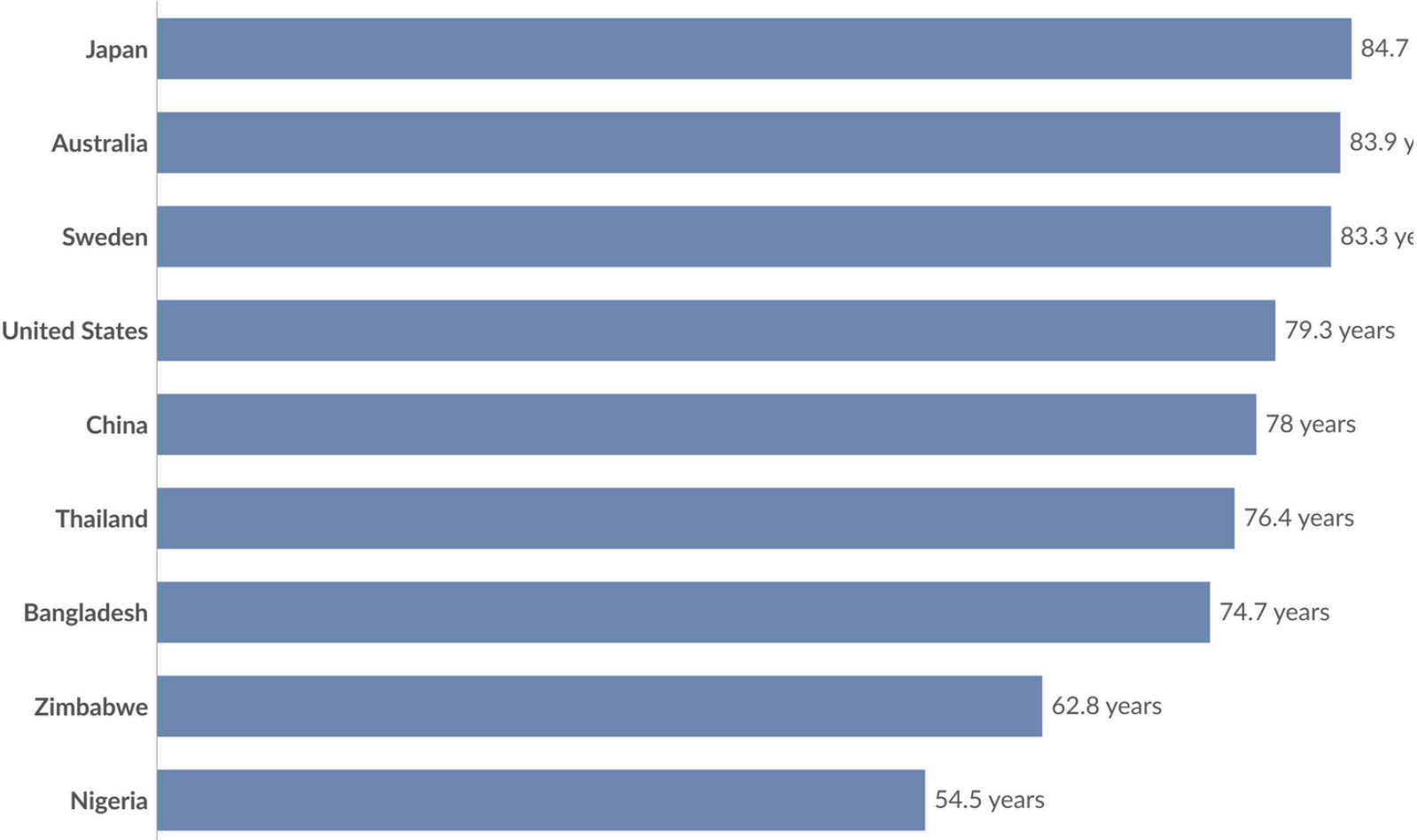
Descriptive statistics

- mean
- median
- variance
- correlations
- conditional expectations

# Life expectancy at birth

## Life expectancy, 2023

The period life expectancy<sup>1</sup> at birth, in a given year.



Life expectancy is based on an estimate of the average age that members of a particular population group will be when they die

Source: [Our World in Data](#)

# Life expectancy by current age

## Life expectancy for people of different ages, Australia, 2023



The total period life expectancy<sup>1</sup> for people who have reached a given age.



Source: [Our World in Data](#)



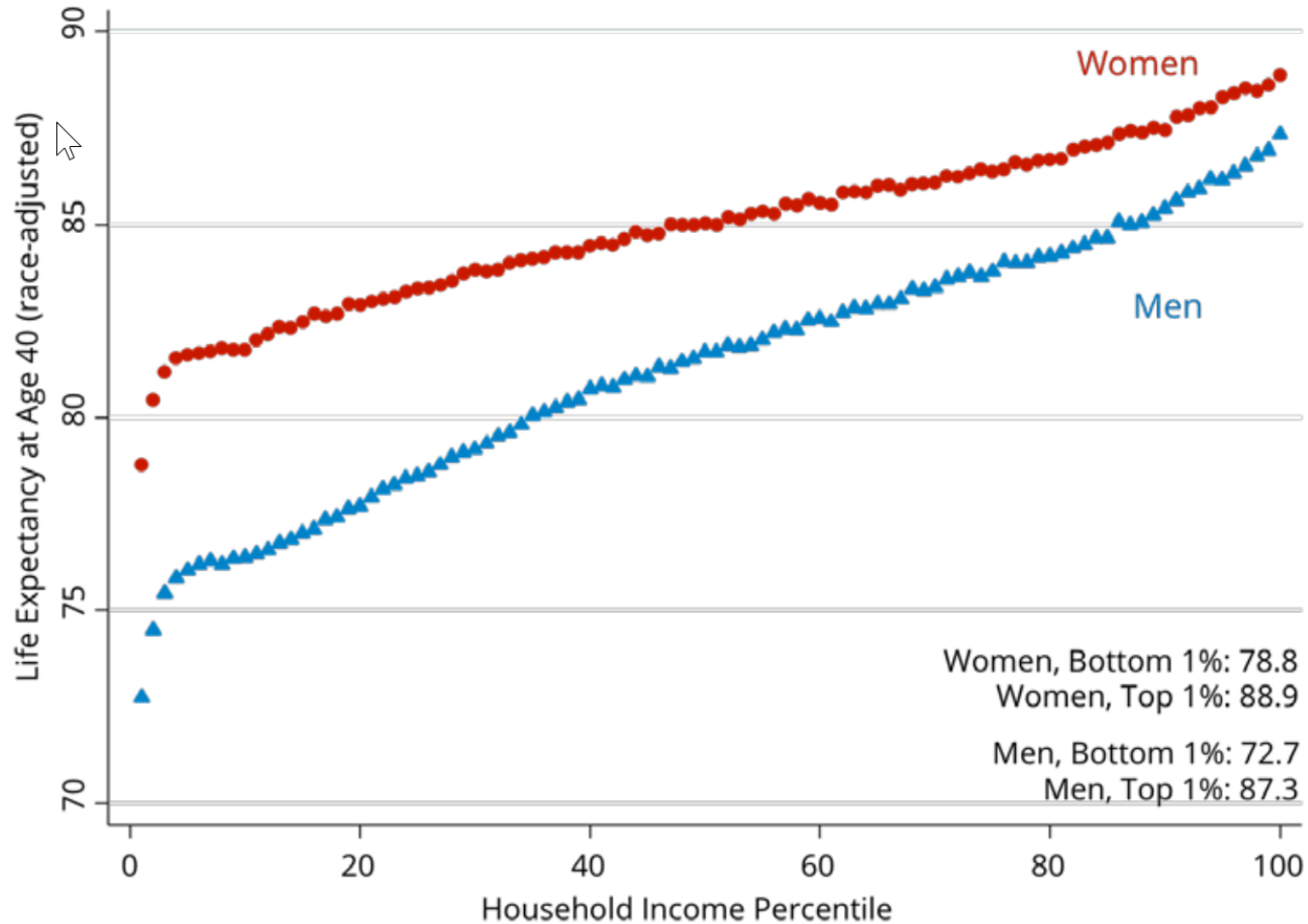
# Guess the gap

US men, aged 40, 2001 to 2014

How many more years does a man in the richest 1% expect to live than a man in the poorest 1%?

Portal round open now: tell your neighbor your number first, then slide

# The gap is 14.6 years



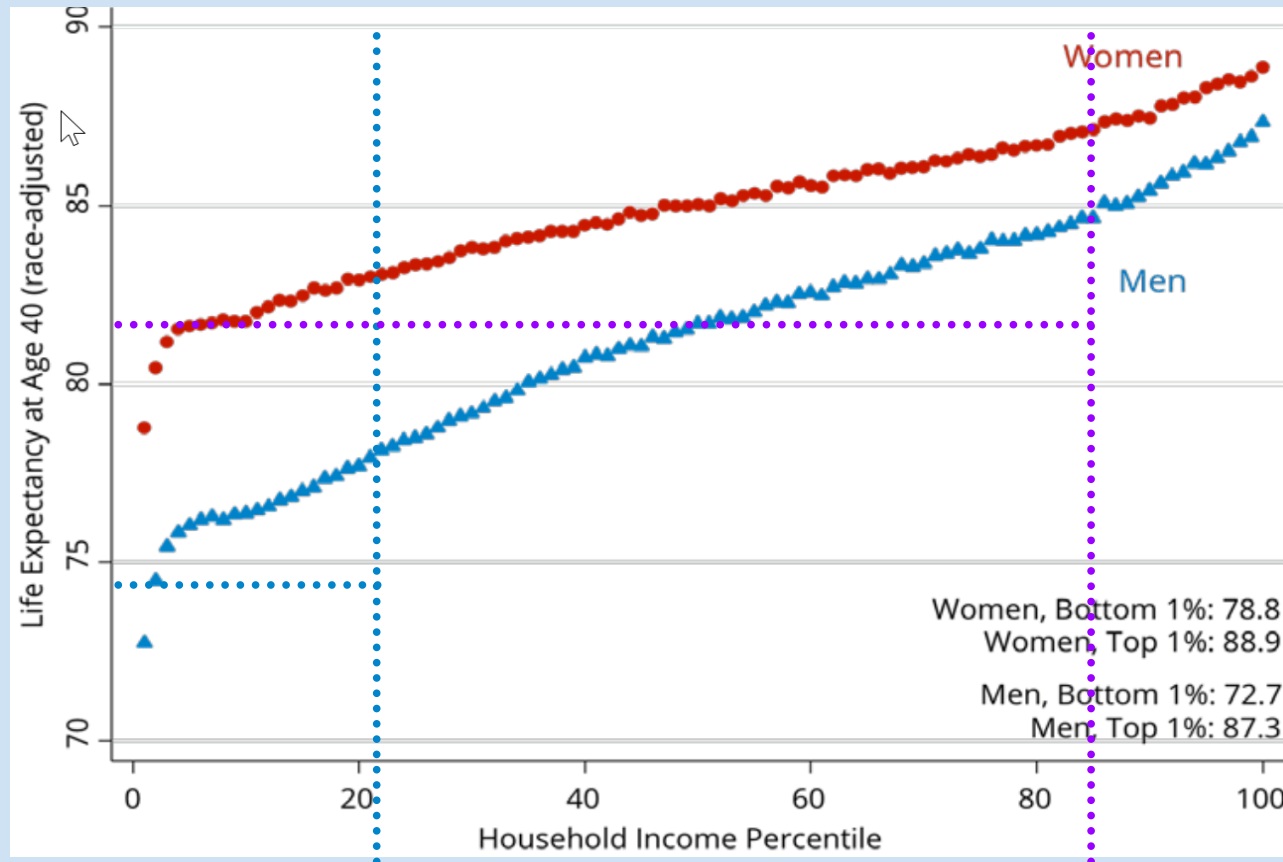
Men, age 40

- Poorest 1%: 72.7 years
- Richest 1%: 87.3 years

Women: a gap of 10.1 years

Source: [Chetty et al. \(2016\)](#)

# Life expectancy in the United States



Y: life expectancy at age 40

X: income percentile

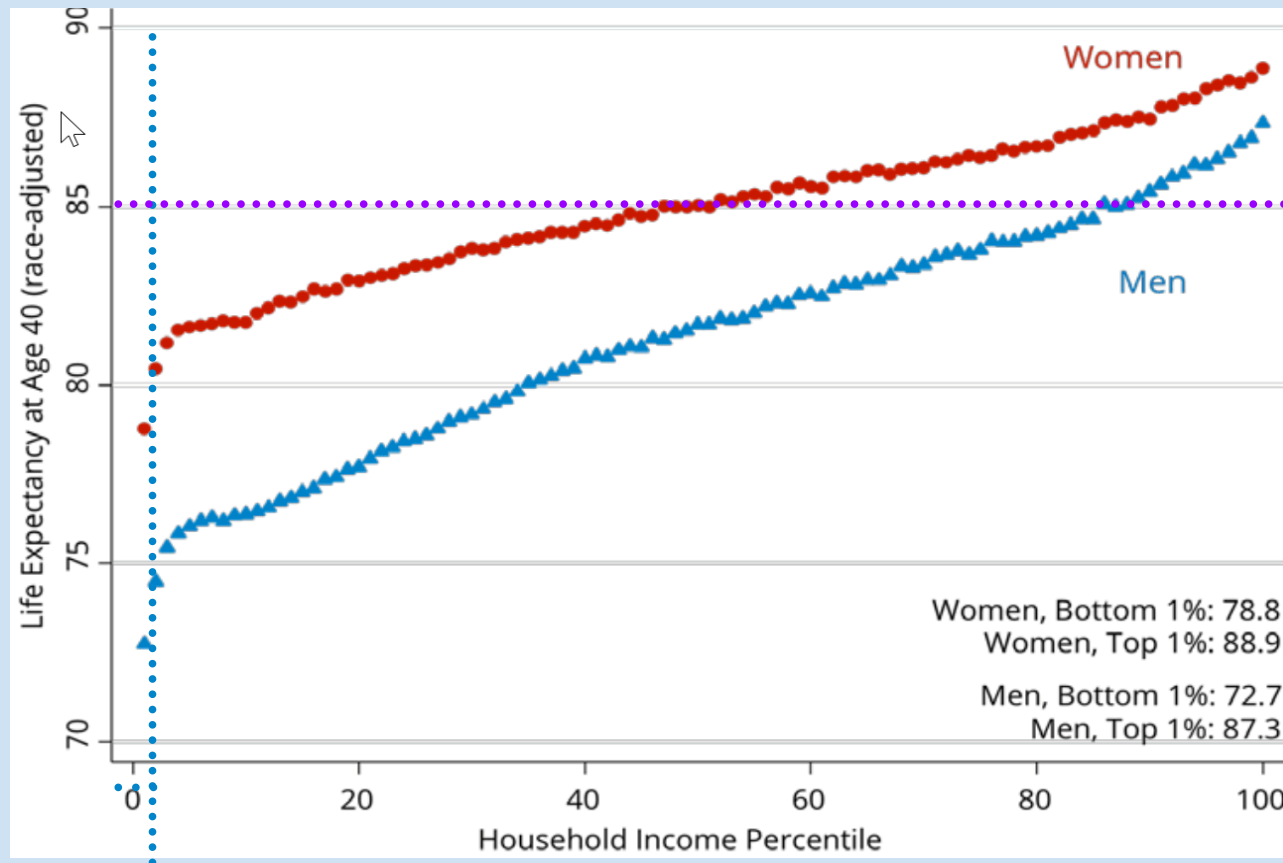
$$E[Y | X = x]$$

Compare:

- $E[Y | X = p20]$
- $E[Y | X = p80]$

Source: [Chetty et al. \(2016\)](#)

# Life expectancy in the United States



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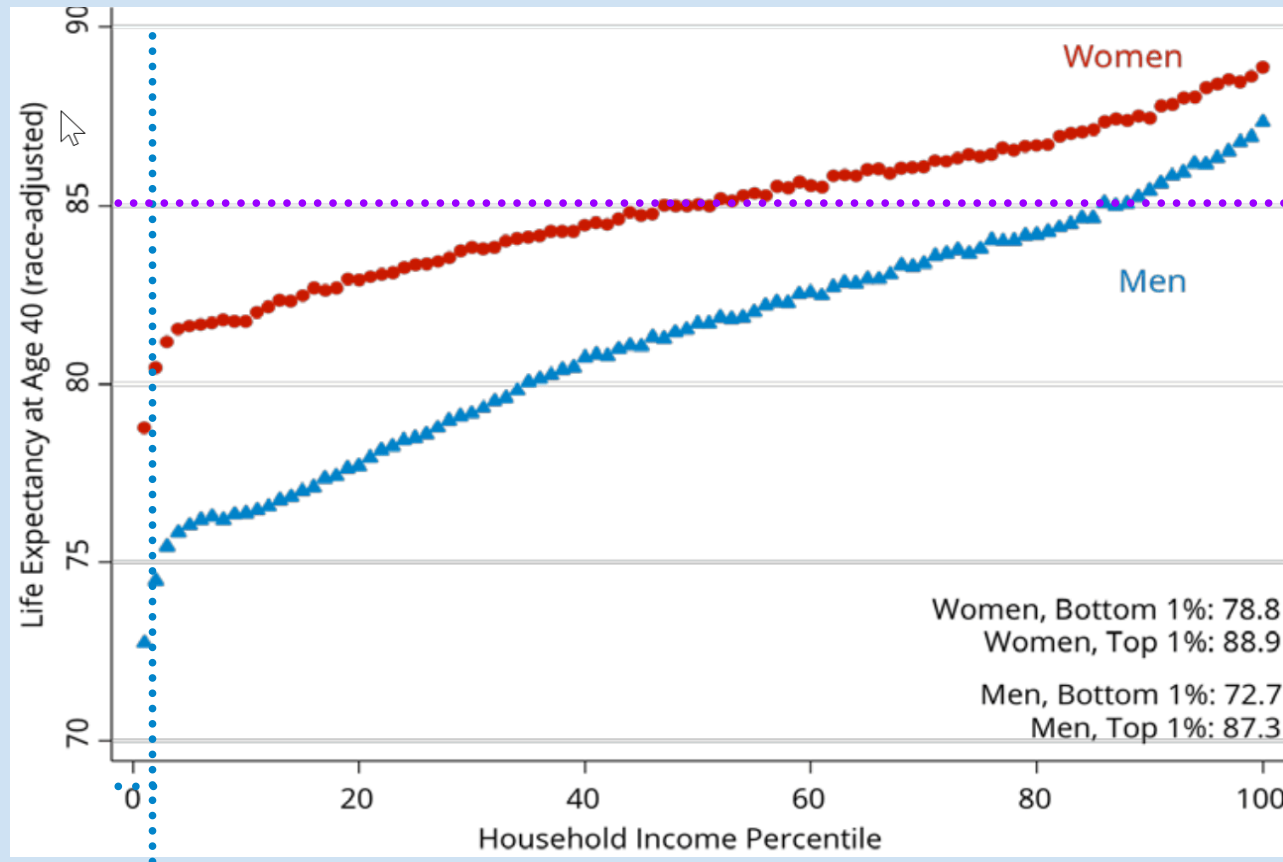
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Compare:

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Source: [Chetty et al. \(2016\)](#)

# Life expectancy in the United States



Y: life expectancy at age 40

X: income percentile

$$E[Y | X = x, Z = z]$$

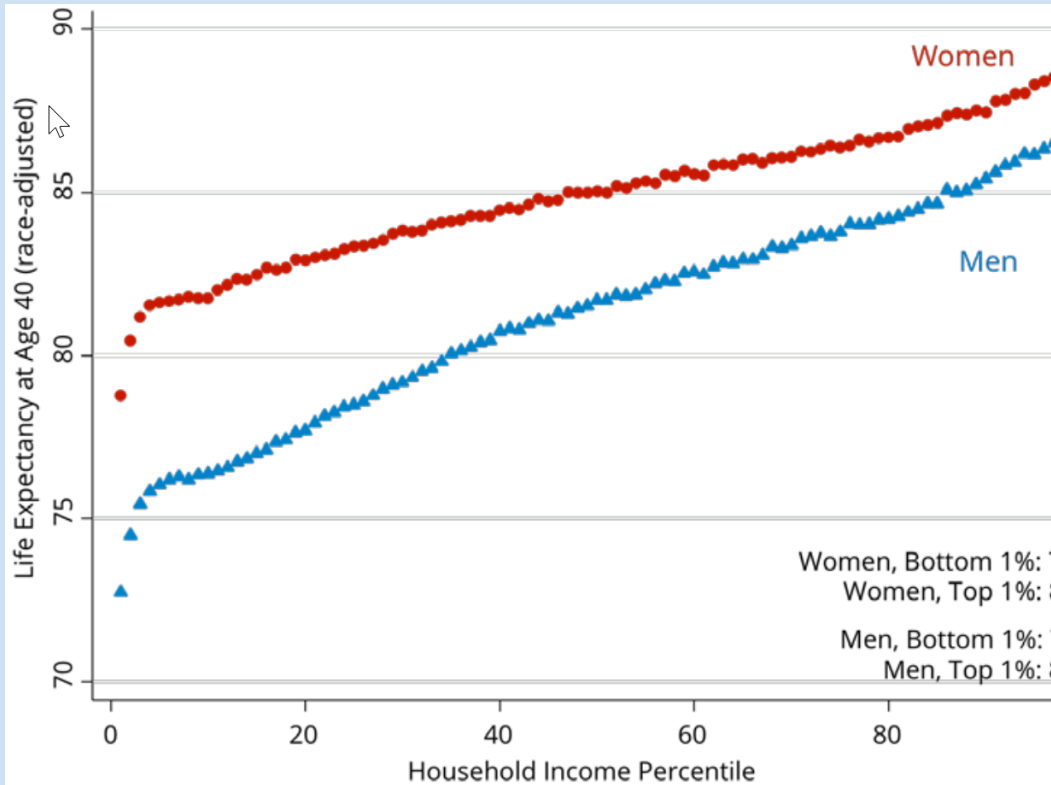
Compare:

- $E[Y | X = p20]$
- $E[Y | X = p80]$

Source: [Chetty et al. \(2016\)](#)

# Unconditional vs. conditional expectation

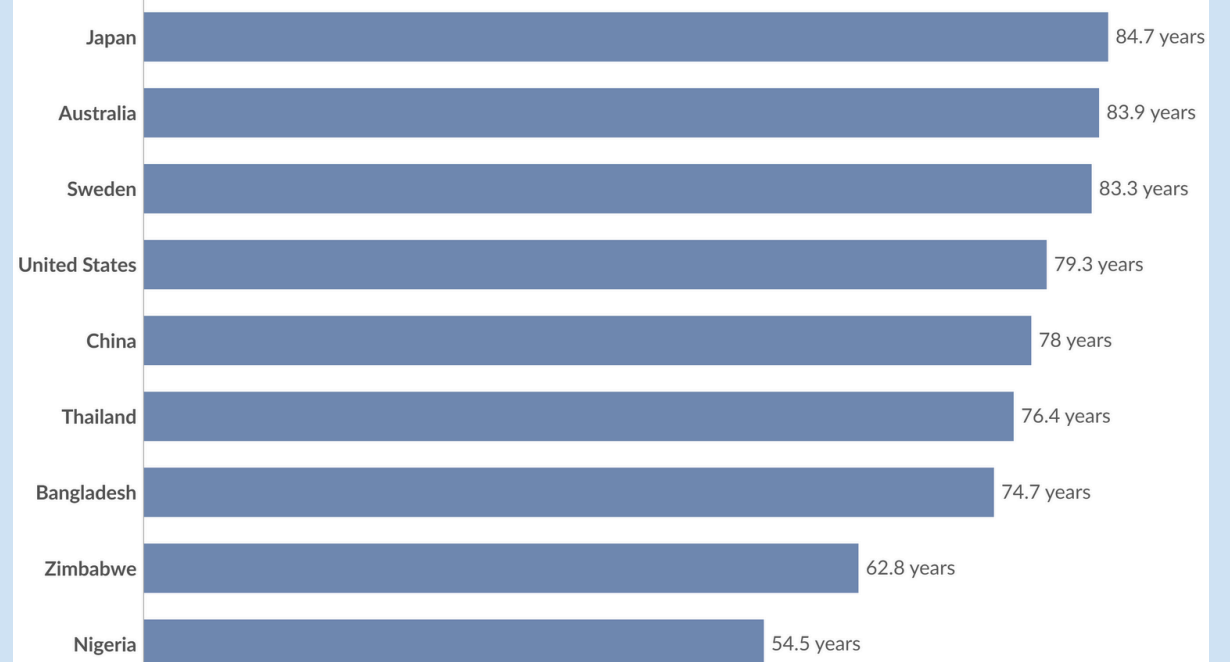
$E[Y]$  hides a lot of heterogeneity that we can see with  $E[Y|X]$ !



## Life expectancy, 2023

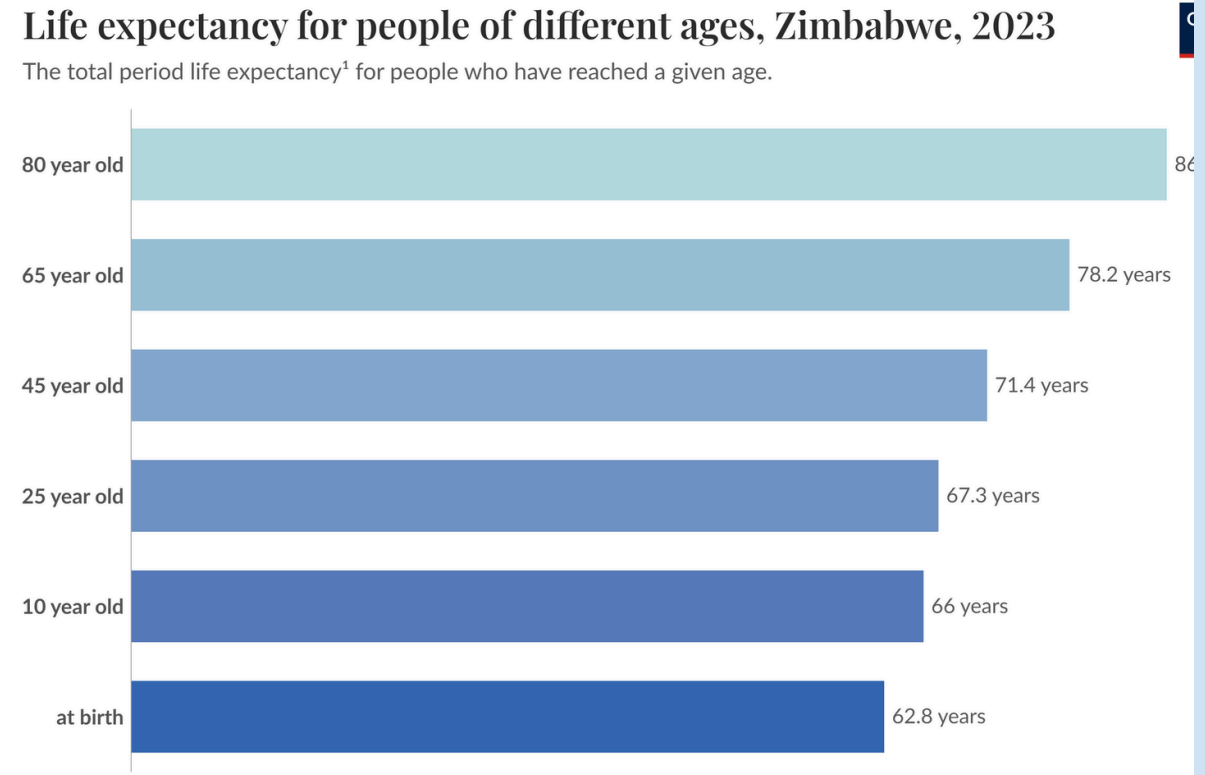
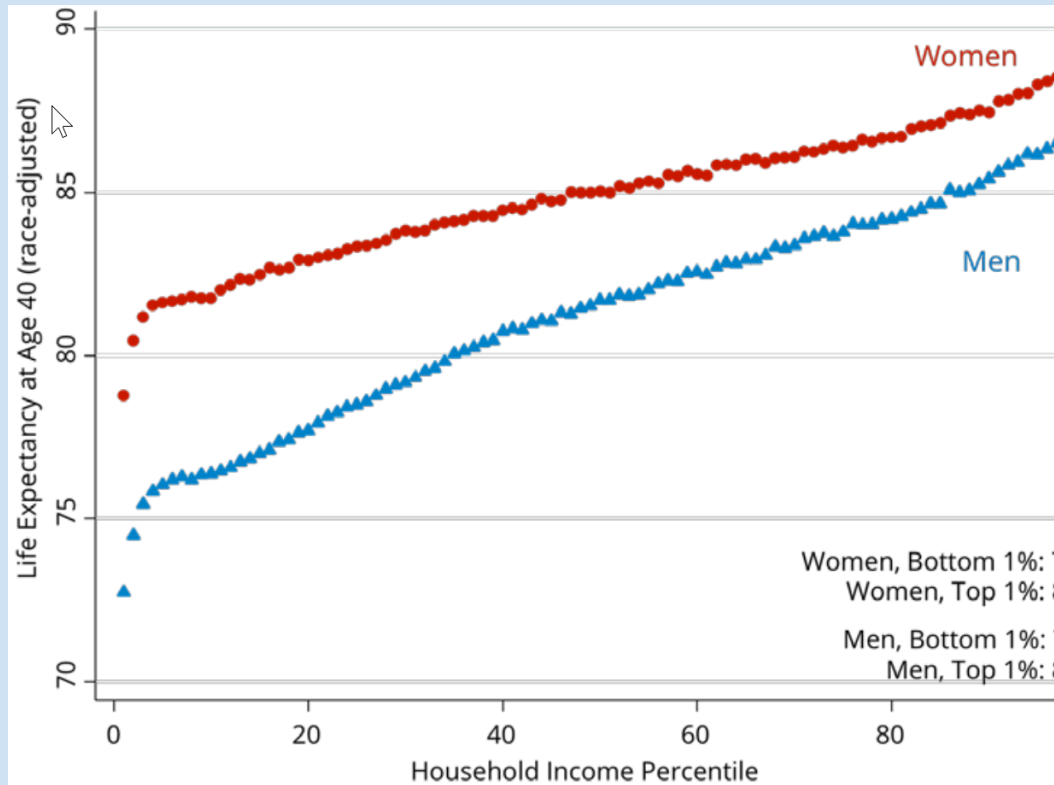
The period life expectancy<sup>1</sup> at birth, in a given year.

Our World  
in Data

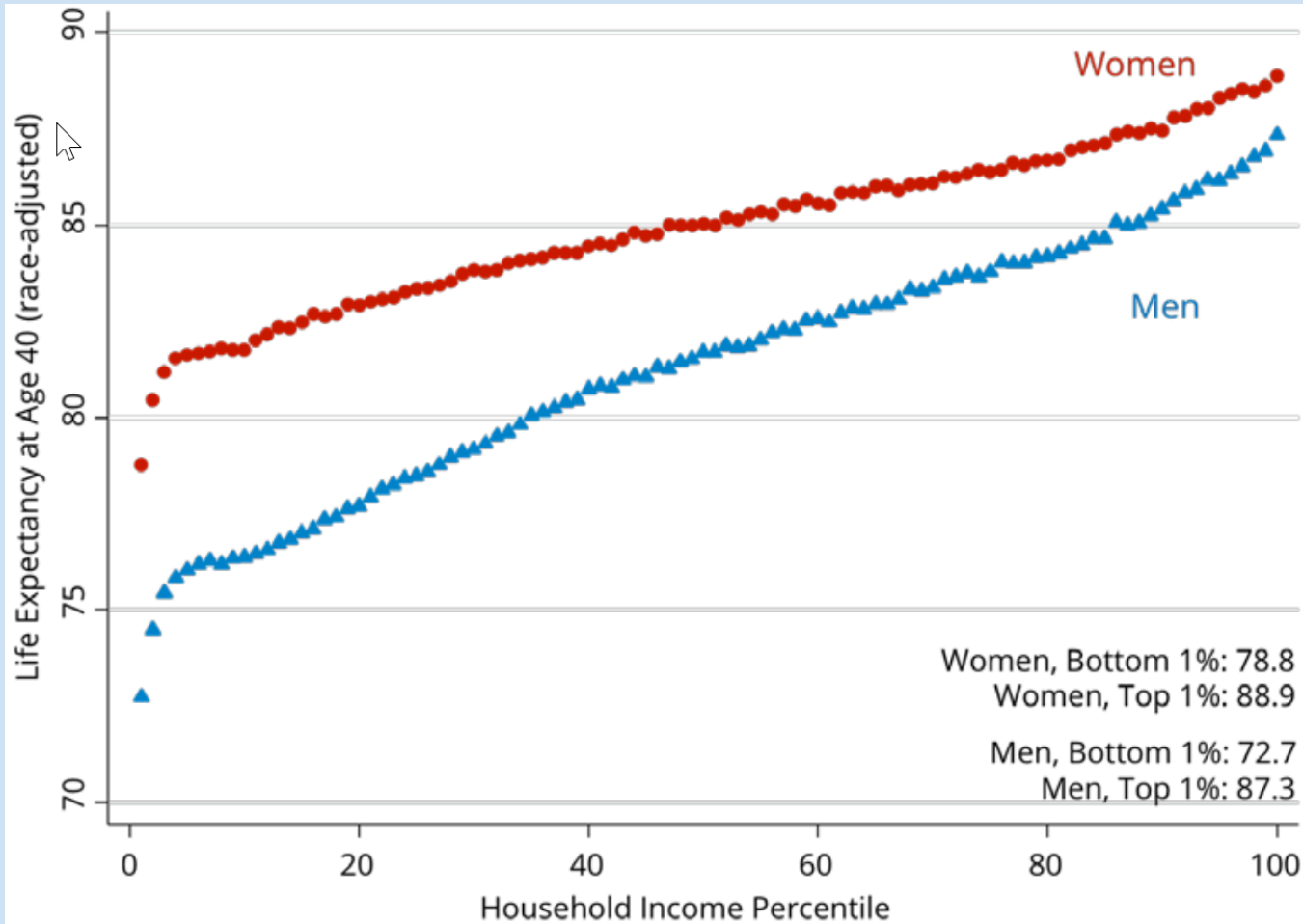


# Importance of looking beyond averages!

Low-income men in the US have  $\approx$  LE as Zimbabwean men!



# Is this a causal relationship?



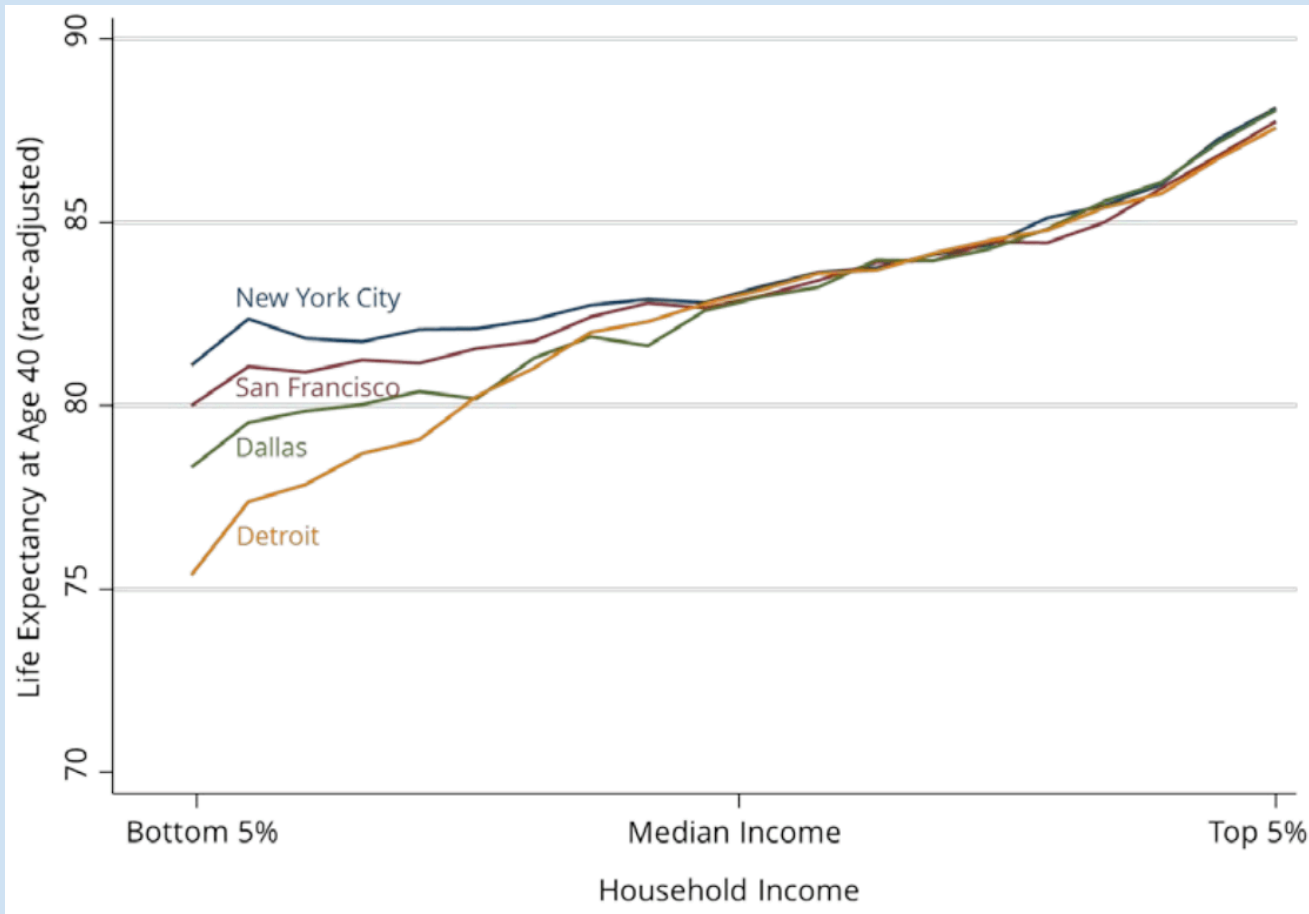
Does this suggest that giving more money to those at the bottom of the income distribution would ↑ their life expectancy?

Source: [Chetty et al. \(2016\)](#)



# Descriptives can be incredibly powerful!

Place matters!



It's better to be poor in New York than in Detroit

Source: [Chetty et al. \(2016\)](#)

Discrete or categorical?

# The recording's rule needs a companion

Essential Concepts, week 3: discrete takes a finite set of values, continuous takes a continuum

But every column in every dataset has finitely many values

- PISA *escs*, the socio-economic index: 62 distinct values in 12,136 rows
- PISA *math* score: 3,858 distinct values

Ask what the variable measures in the world, not how many values the column has

# Three kinds of variables in the world

| In the world  | Example                                  | Arithmetic makes sense?                          |
|---------------|------------------------------------------|--------------------------------------------------|
| A count       | number of siblings: 0, 1, 2, ...         | yes: an average of 1.4 siblings                  |
| A measurement | math score, socio-economic index, height | yes: any value in a range, rounded when recorded |
| A label       | degree enrolled in, suburb, gender       | no: there is no average suburb                   |

# Codes look numeric, and Python averages them

Education recorded as 1 = completed primary, 2 = completed high school, 3 = completed a degree

`df['education'].mean()` returns 2.3

2.3 is a fact about the codes, not about anyone's education

# Rounding keeps arithmetic, binning kills it

## Rounded

- A score of 61.4 recorded as 61
- Average of 61, 74, 58: 64.3, and it means something

## Binned

- PISA [book](#): books at home, recorded as 0-10, 11-25, 26-100, 101-200, 201-500, 500+
- Average of "101-200" and "26-100": nothing

Books at home is a count in the world; the survey turned it into a label at collection time

# Two words for the label case, both in use

- The Essential Concepts recordings say discrete for anything that is not continuous
- Categorical is the everyday word for the label kind: it names why averaging is nonsense
- You will meet both; when a variable is a label, say categorical and nobody will object

# The correlation gallery

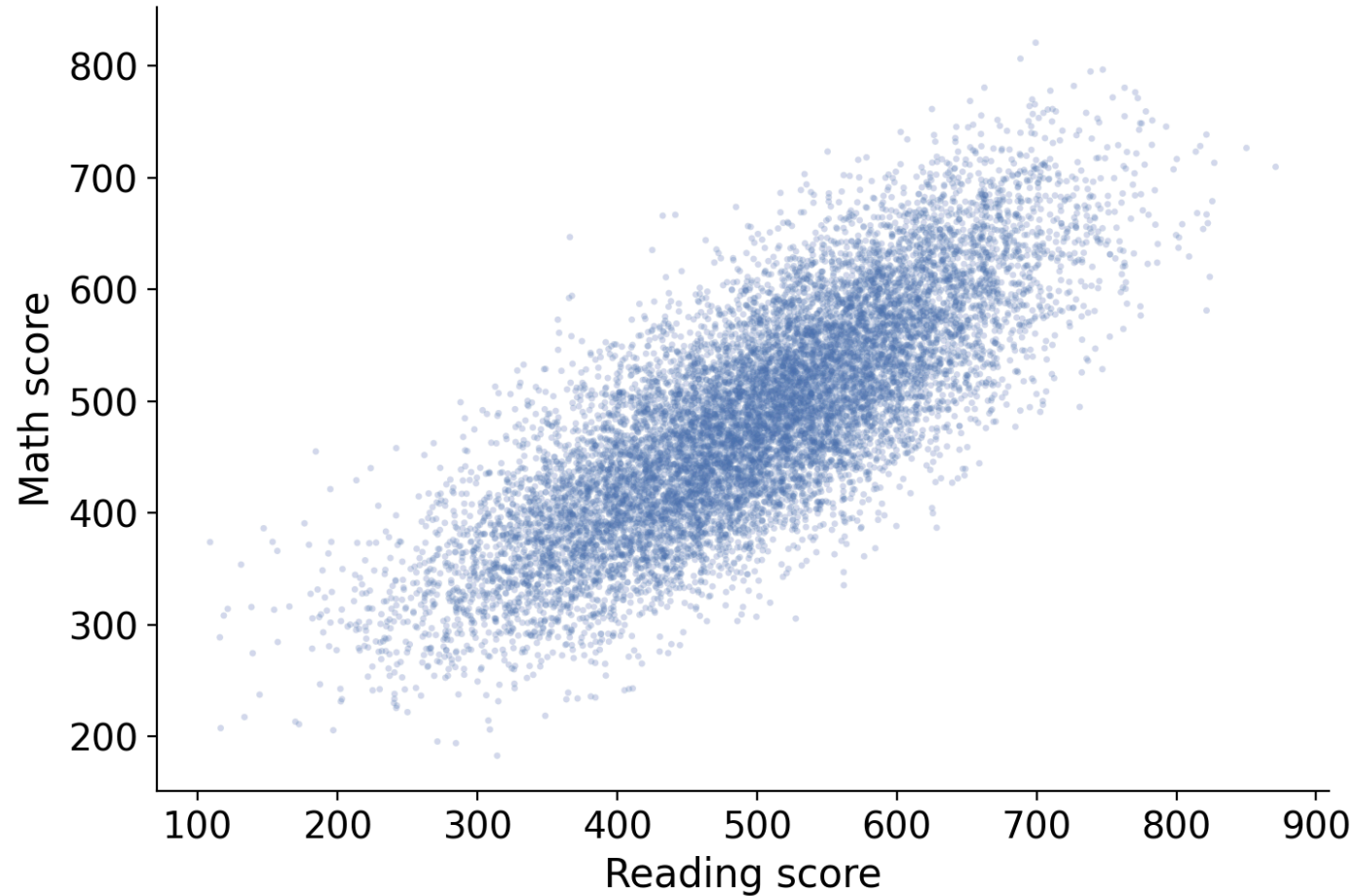
## Six clouds of dots

For each one: tell your neighbor your guess, then slide to your own answer on the portal

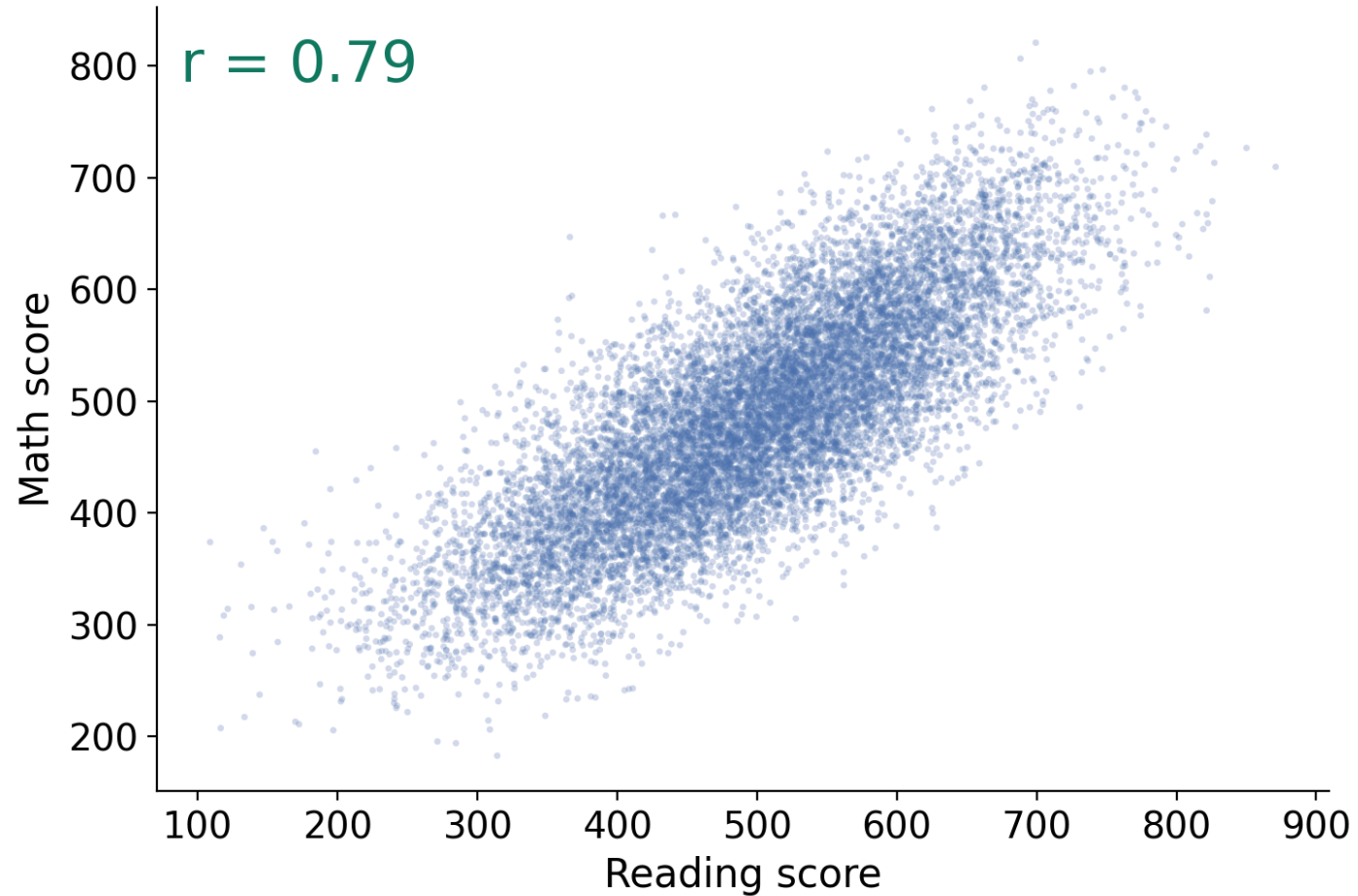
Then we see the room's guesses against the truth



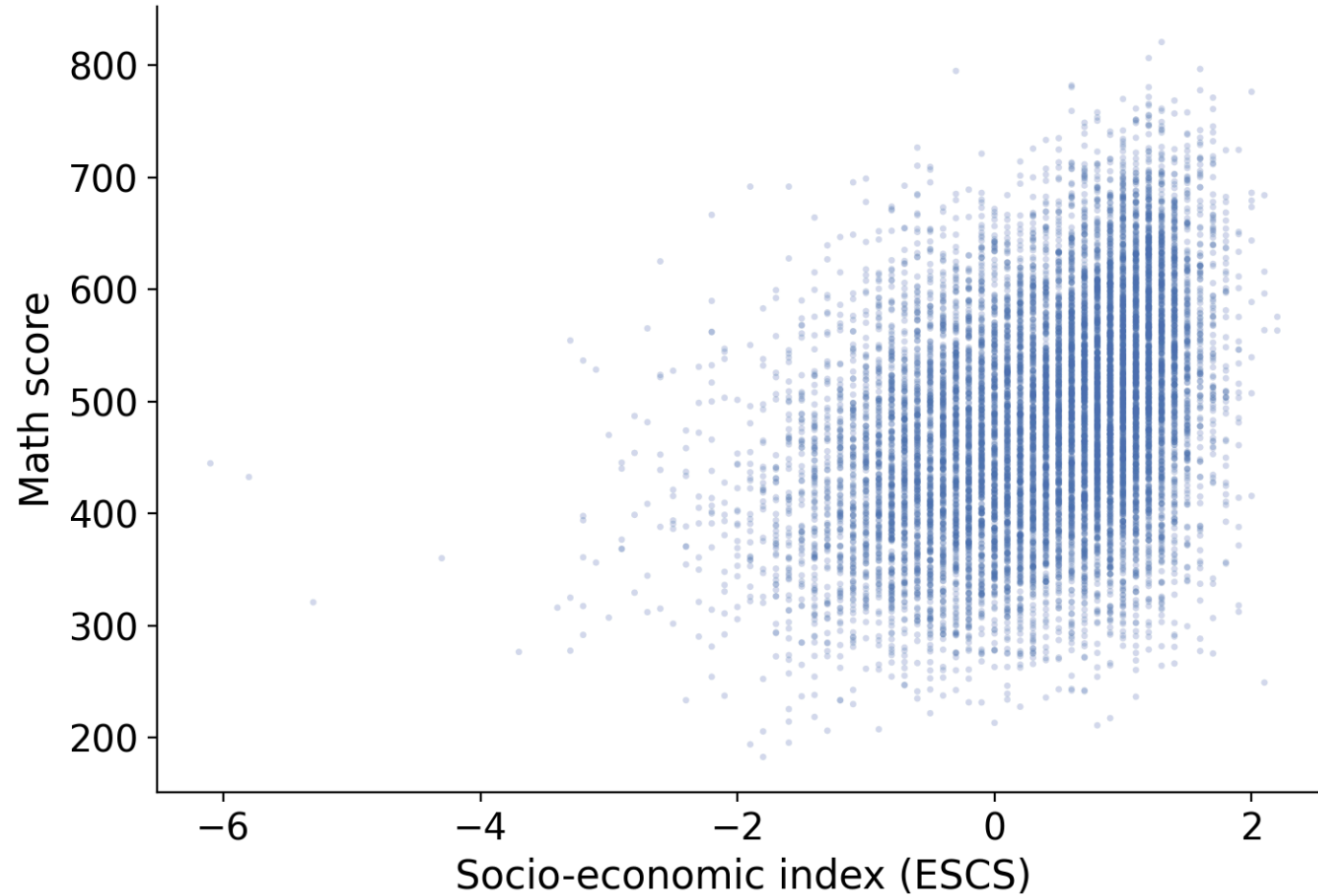
# Gallery 1: math and reading, 12,136 students



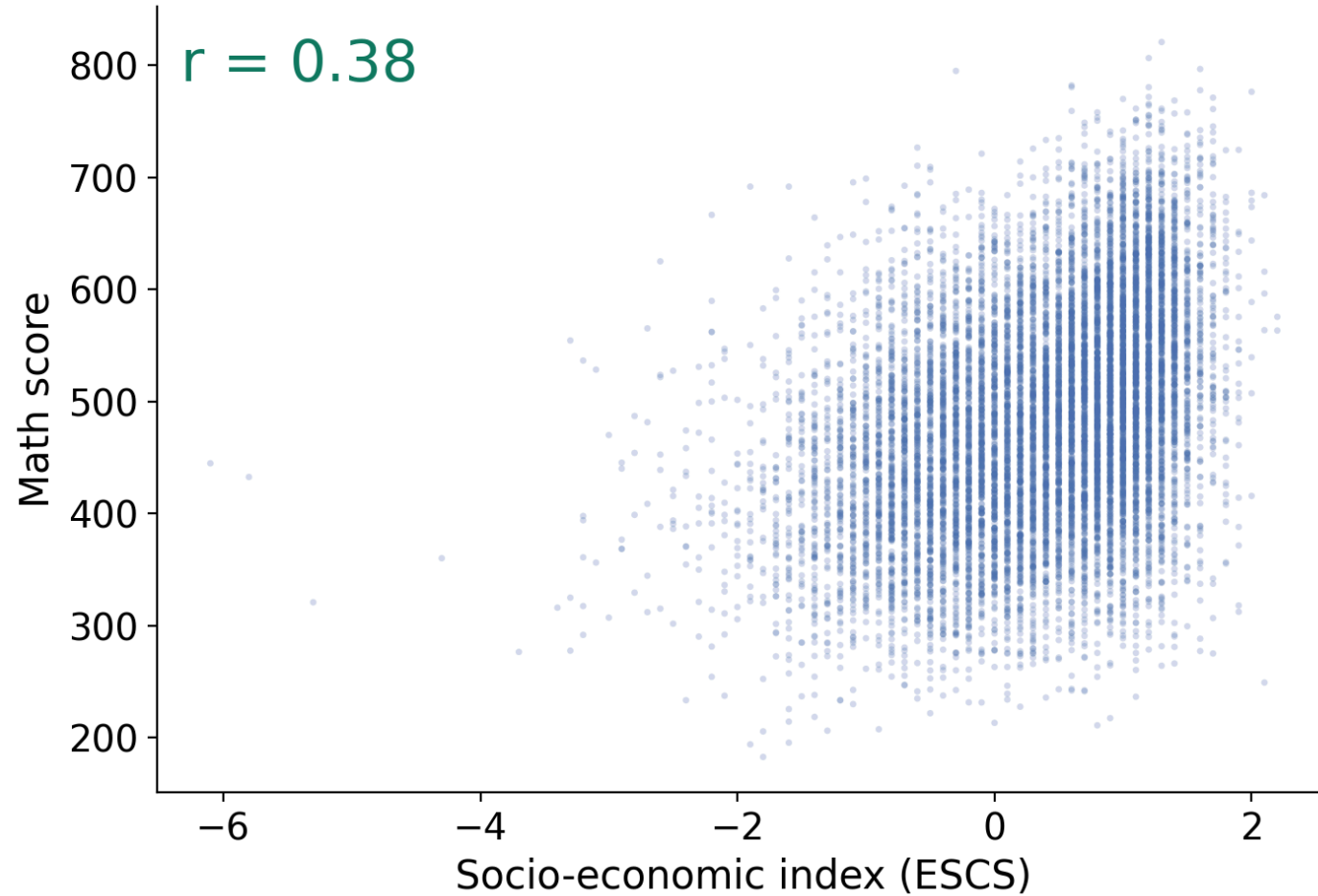
# Gallery 1: math and reading, $r = 0.79$



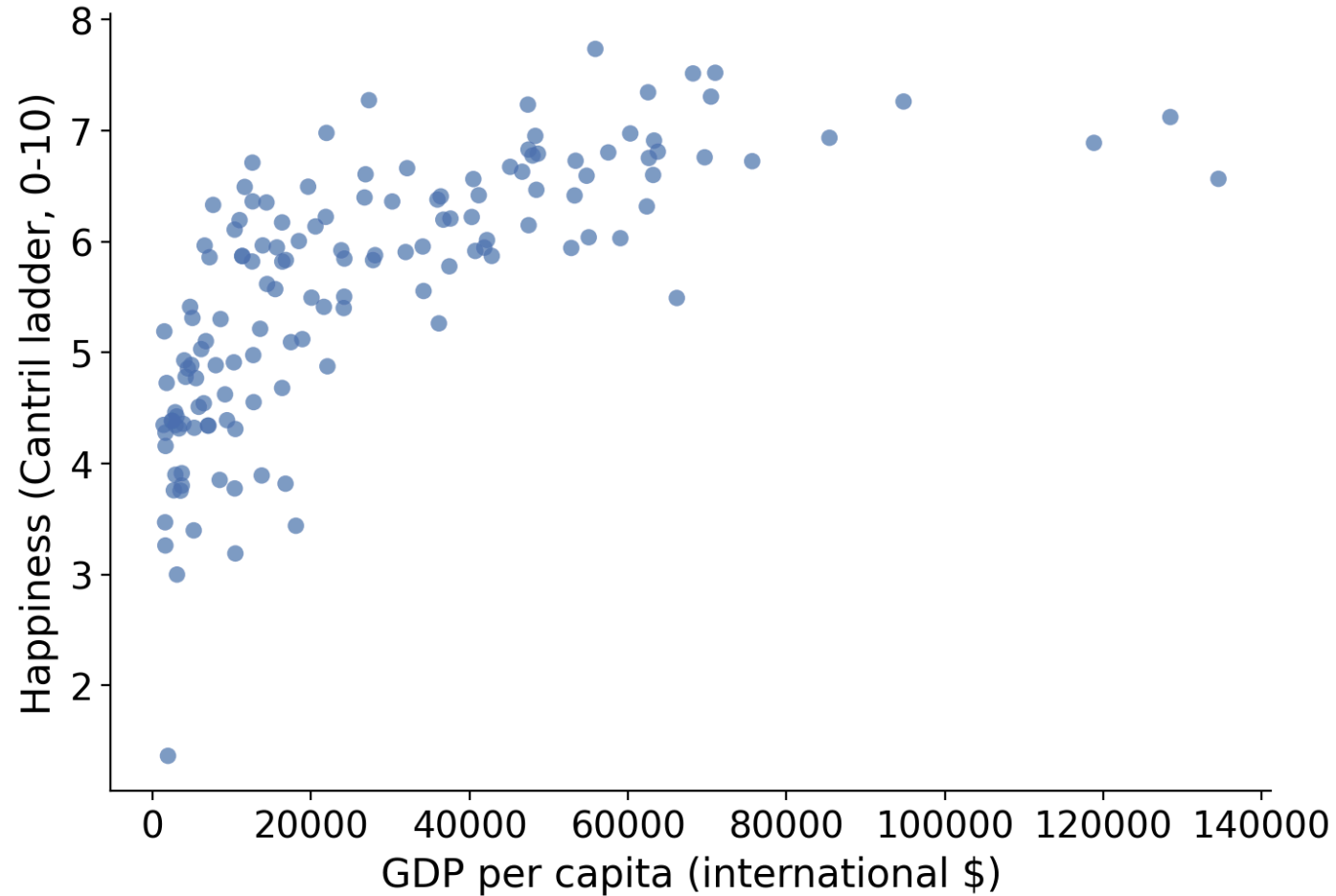
# Gallery 2: math and family background



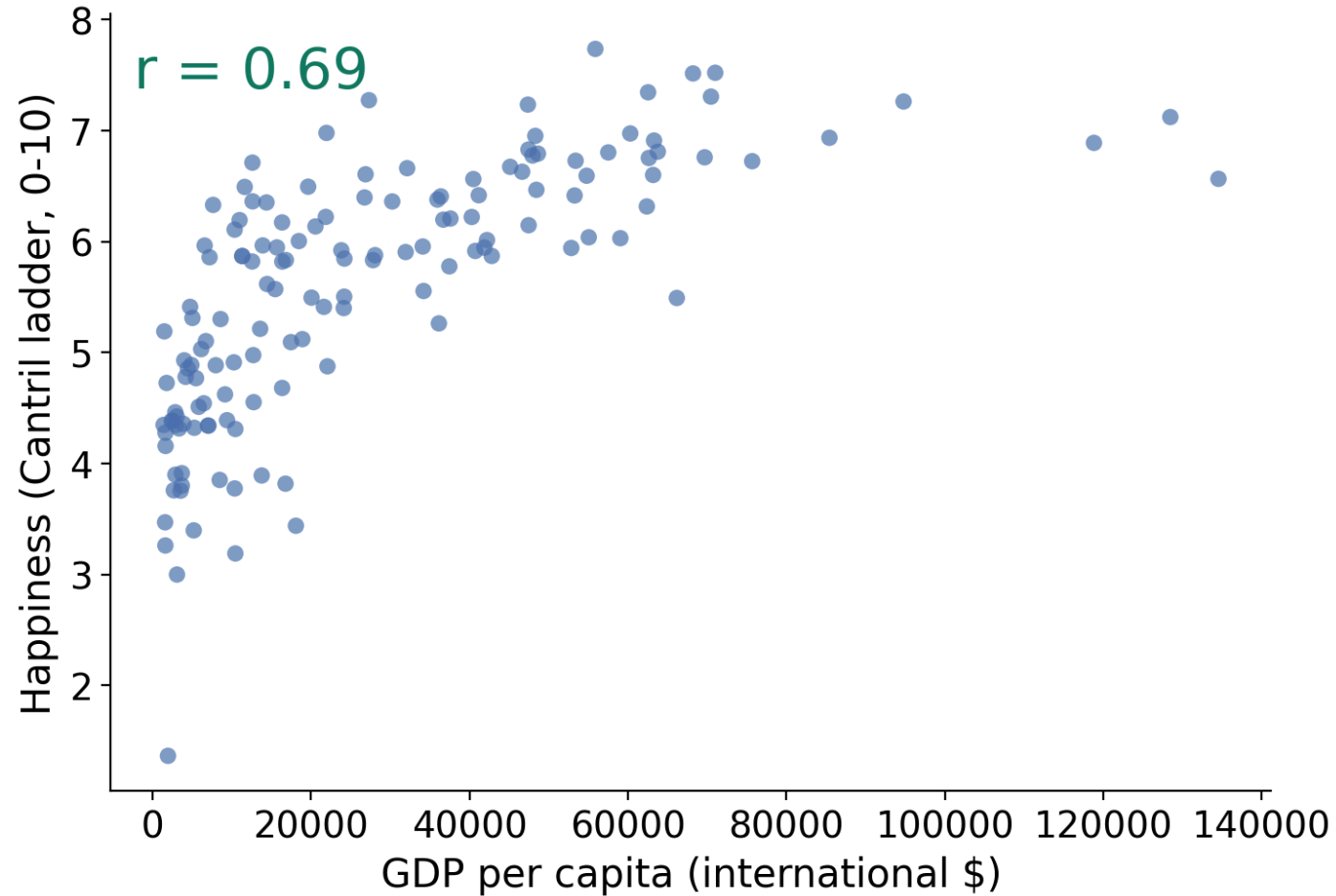
# Gallery 2: math and family background, 0.38



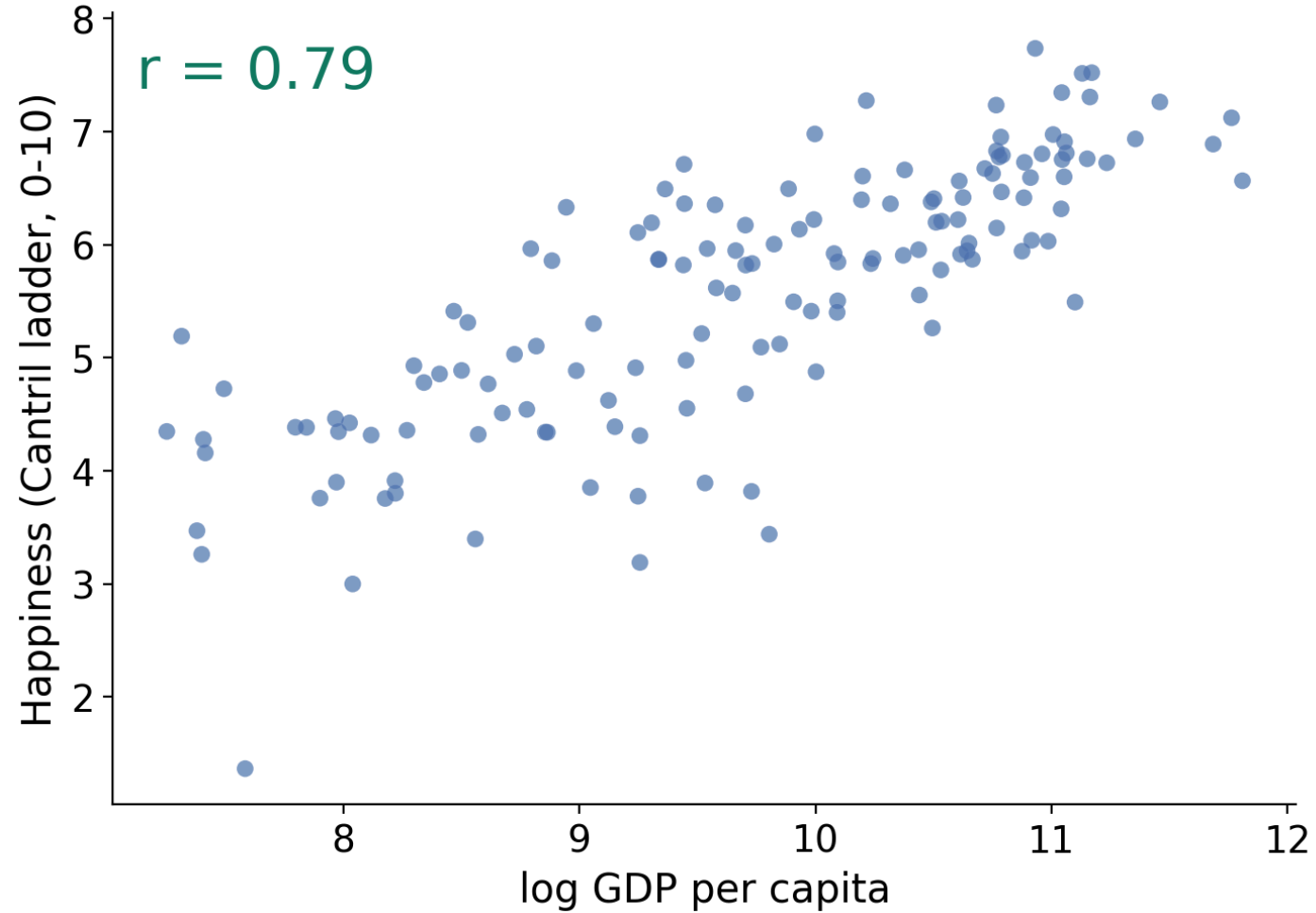
# Gallery 3: happiness and GDP, 144 countries



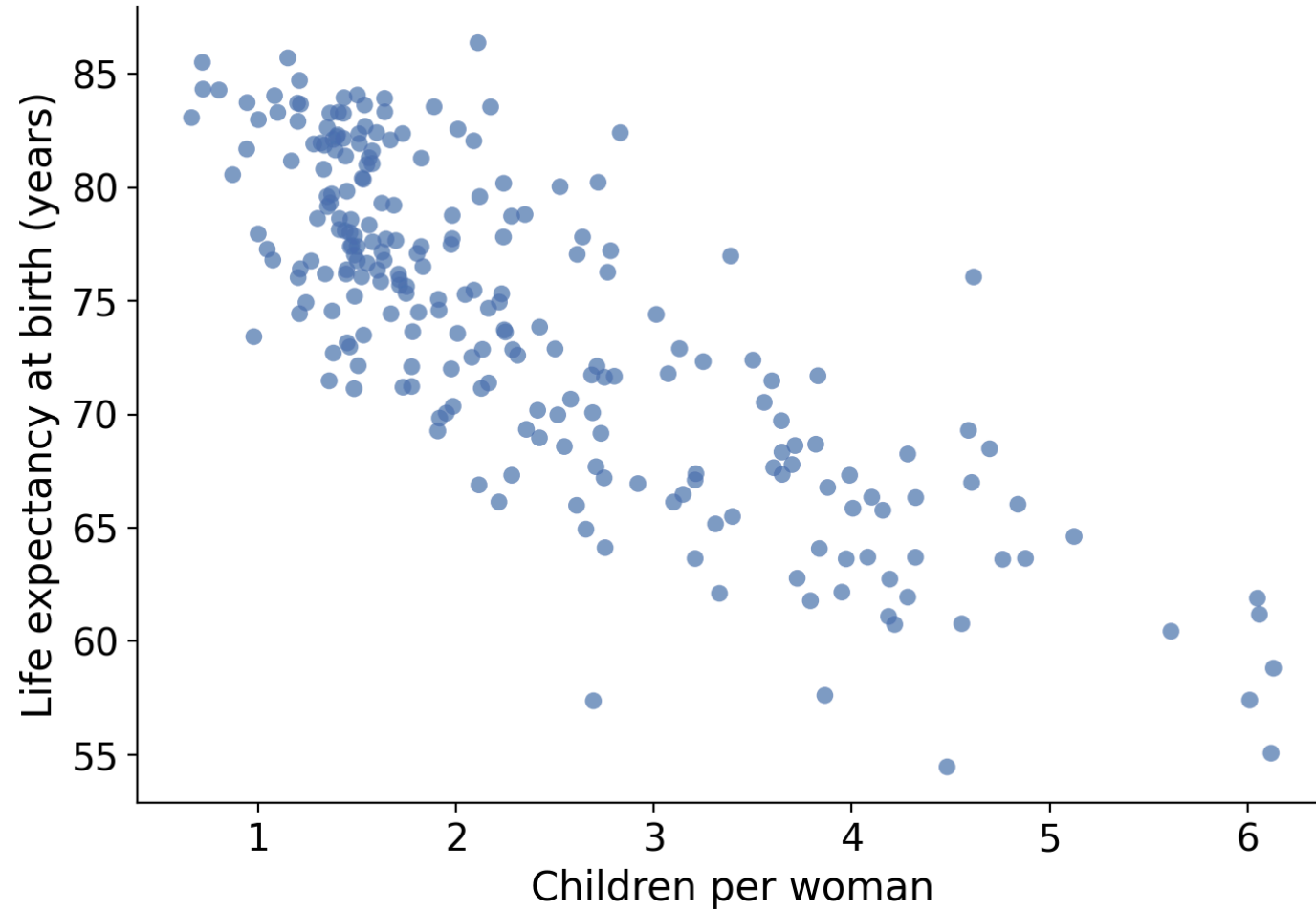
# Gallery 3: happiness and GDP, $r = 0.69$



# Same countries, log GDP: $r = 0.79$

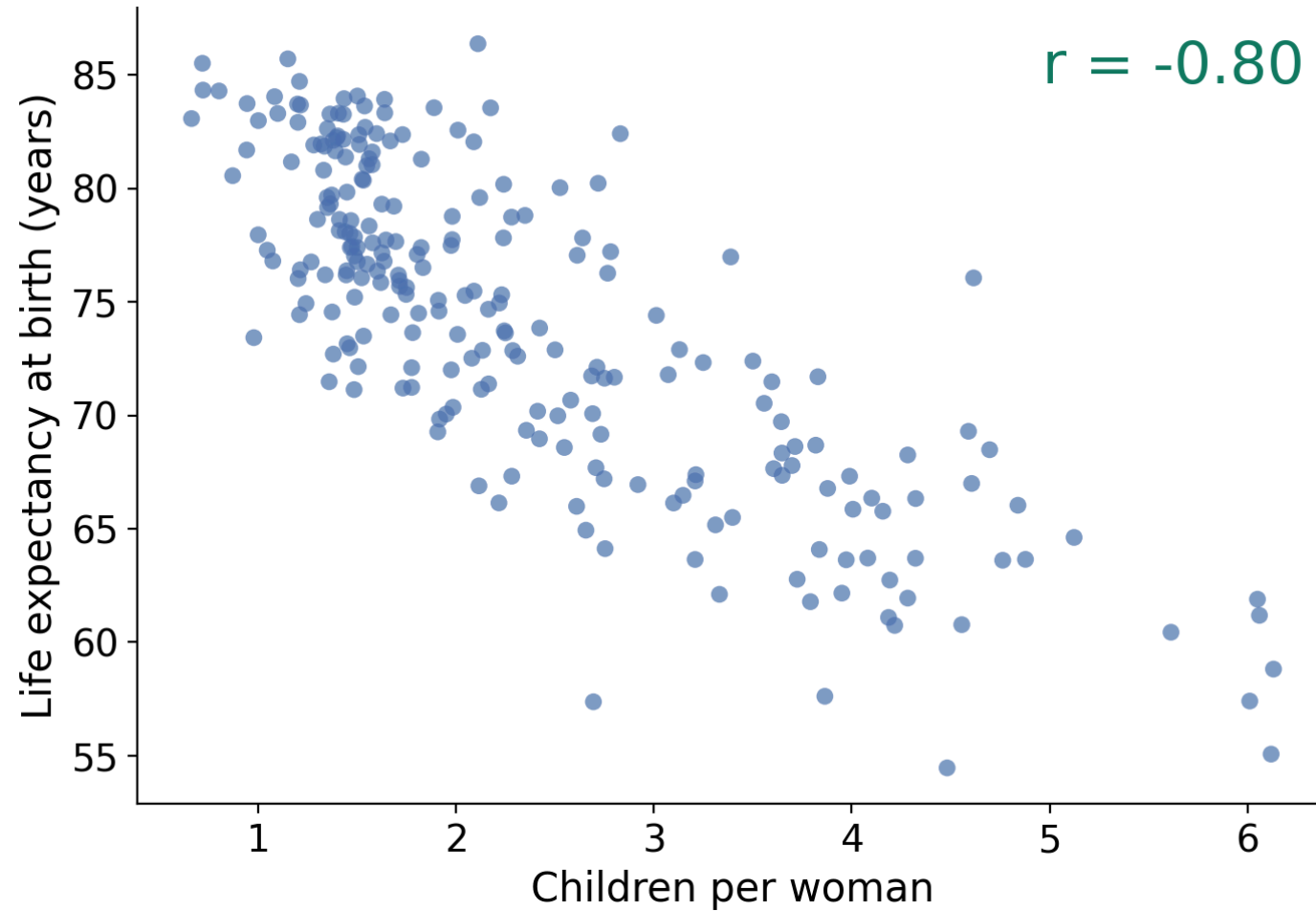


# Gallery 4: fertility and life expectancy

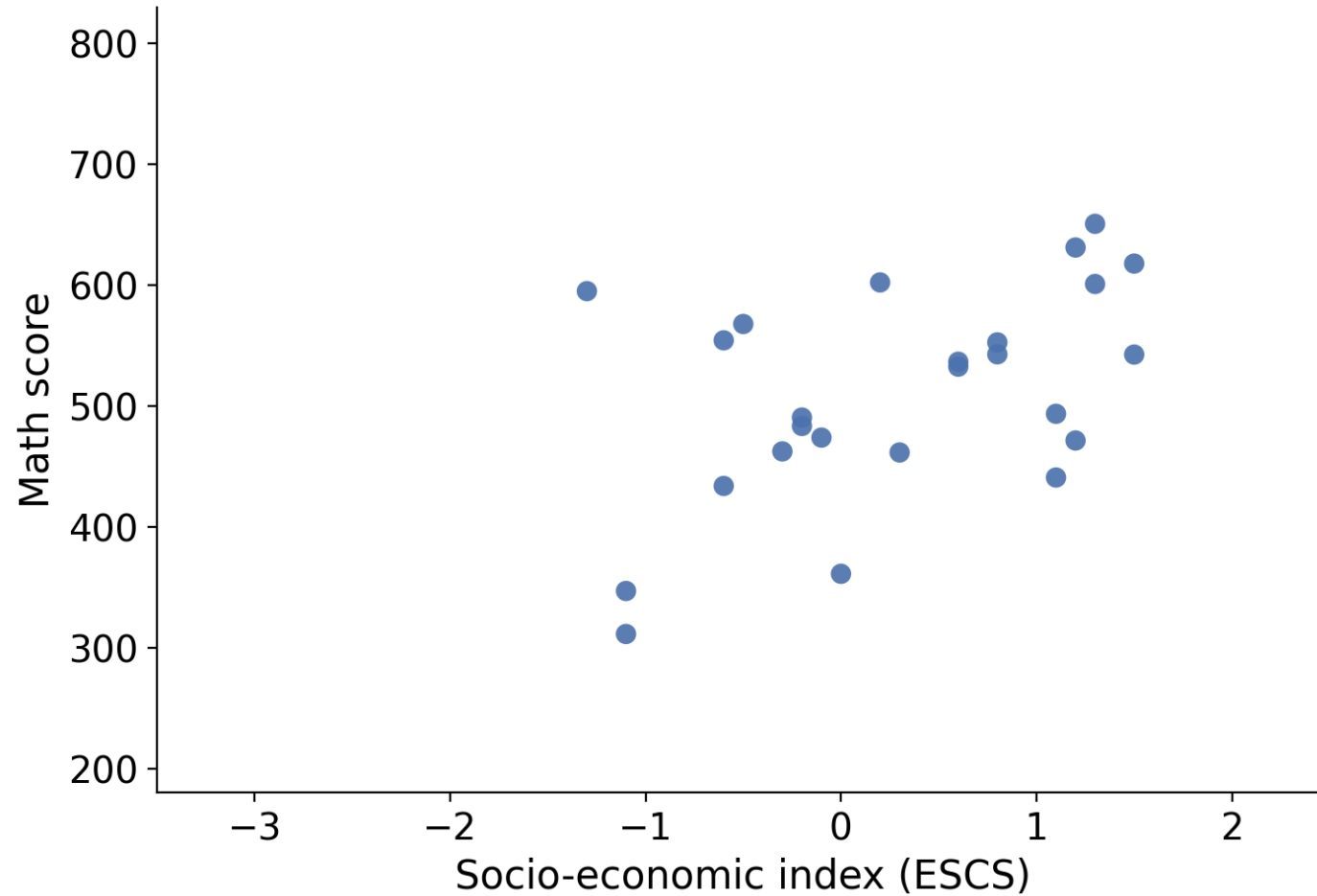




# Gallery 4: fertility and life expectancy, $-0.80$

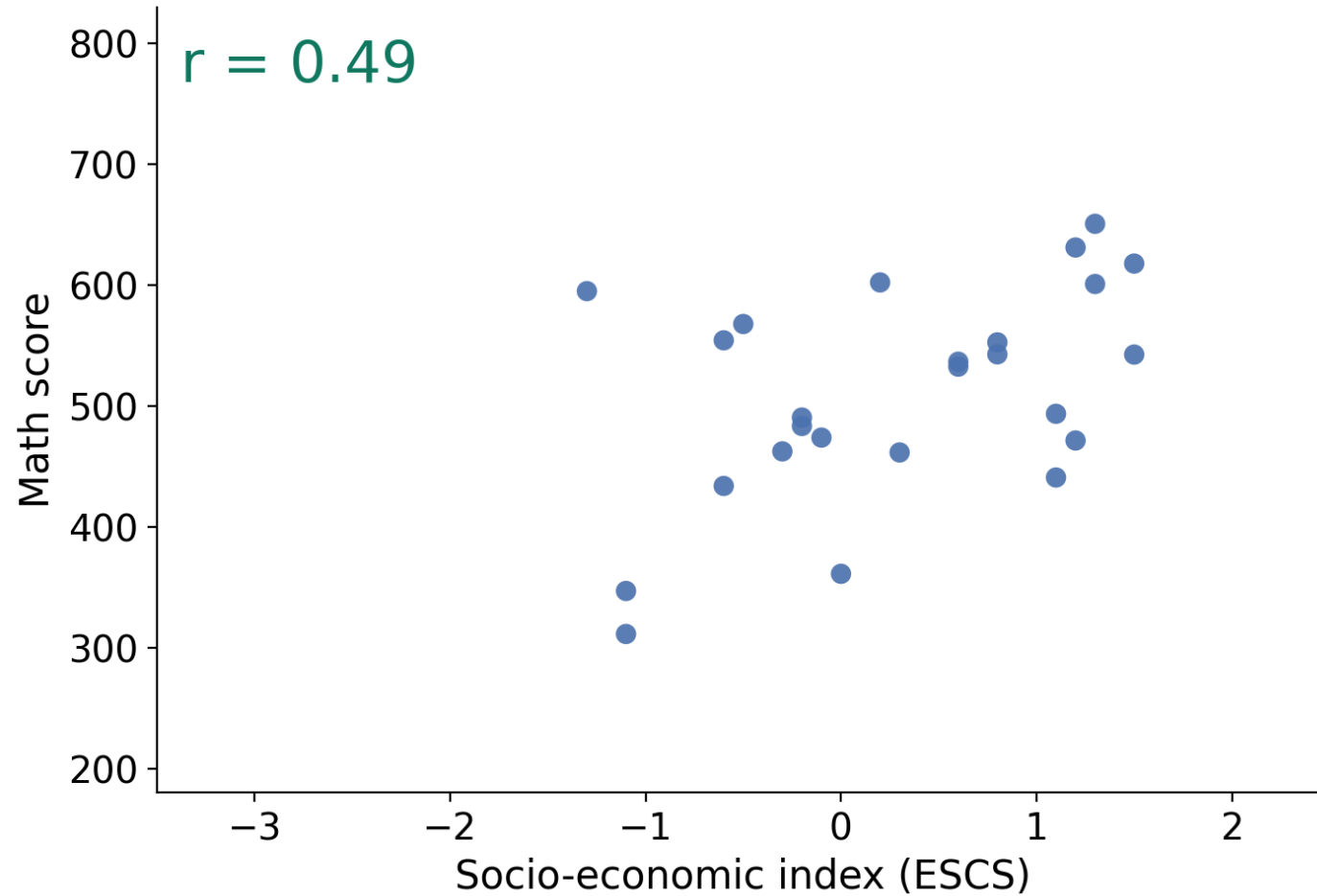


# Gallery 5: the same pair, 25 students



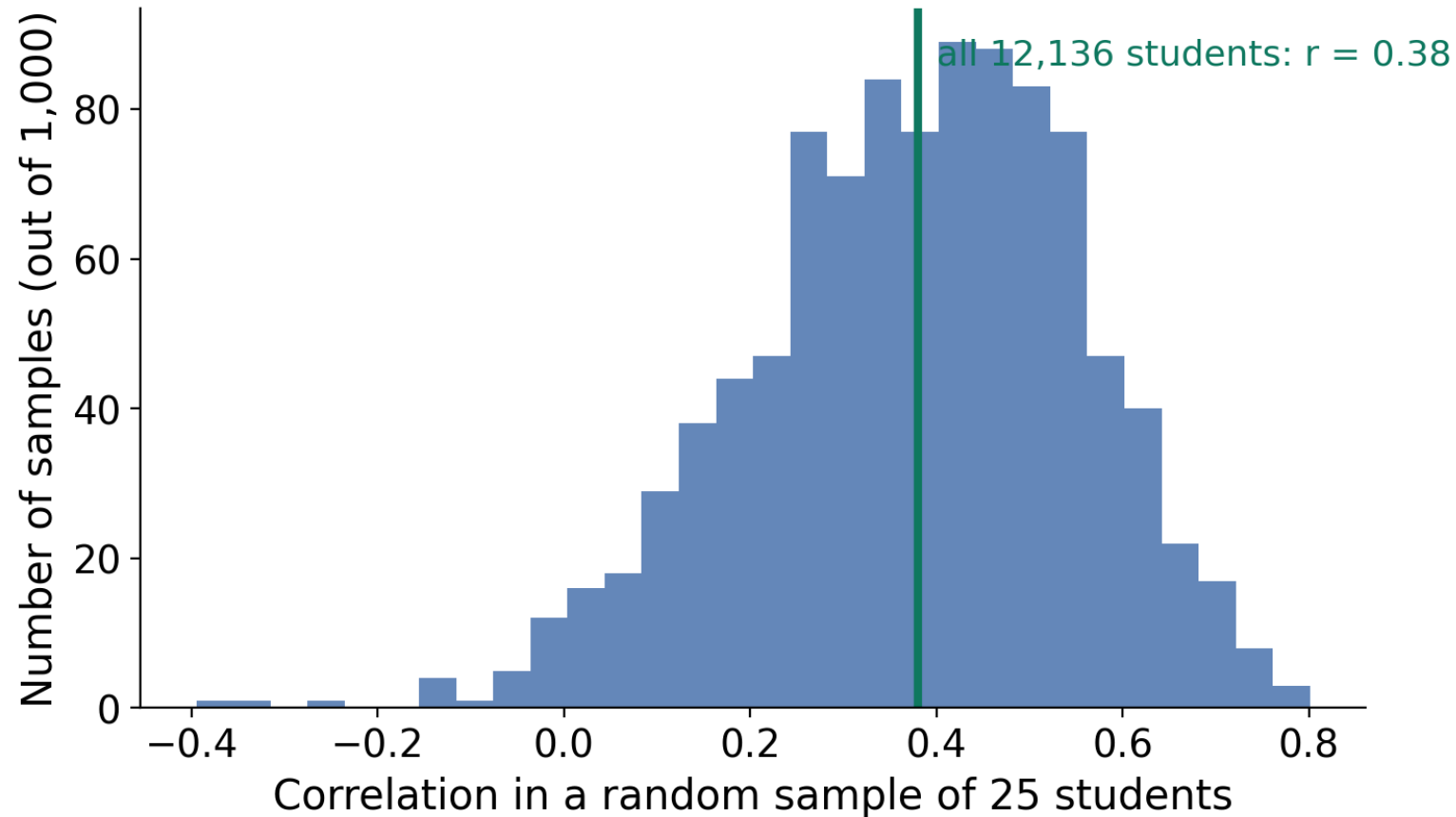
Data: PISA 2022, Australia (OECD), 25 students drawn at random

# Gallery 5: 25 students, $r = 0.49$

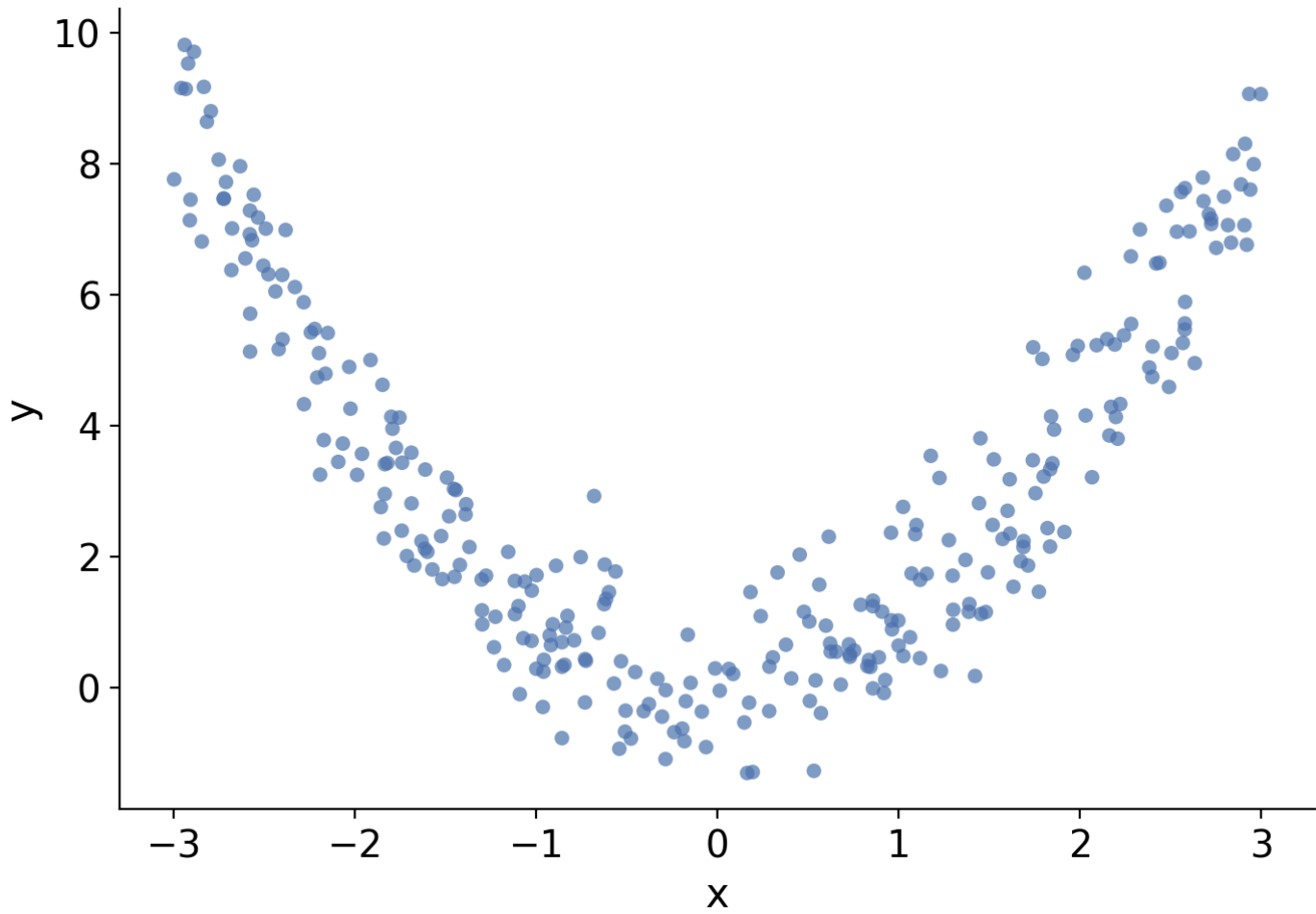


Data: PISA 2022, Australia (OECD), 25 students drawn at random

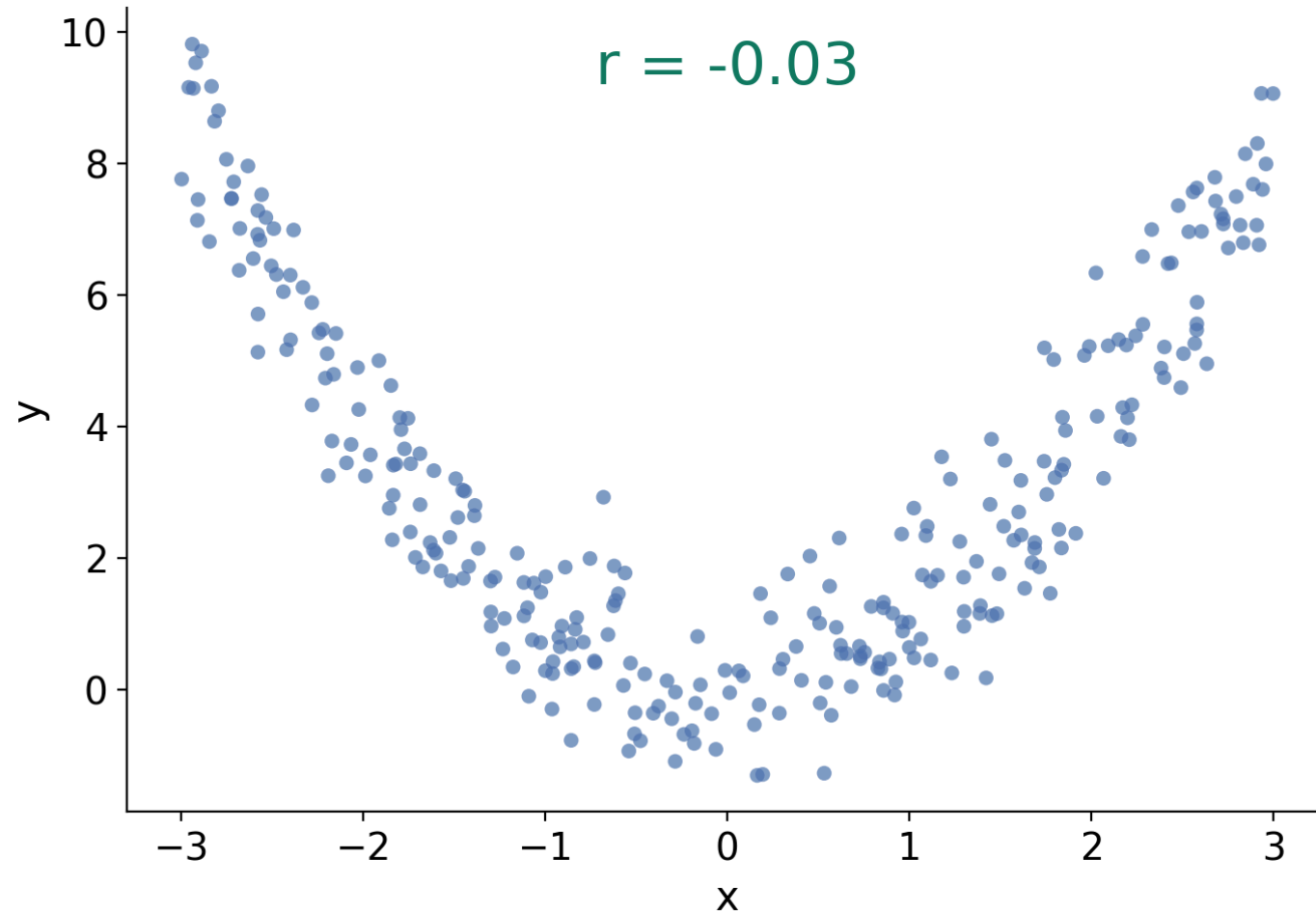
# A thousand samples of 25: $r$ from $-0.4$ to $0.8$



# Gallery 6: one last cloud

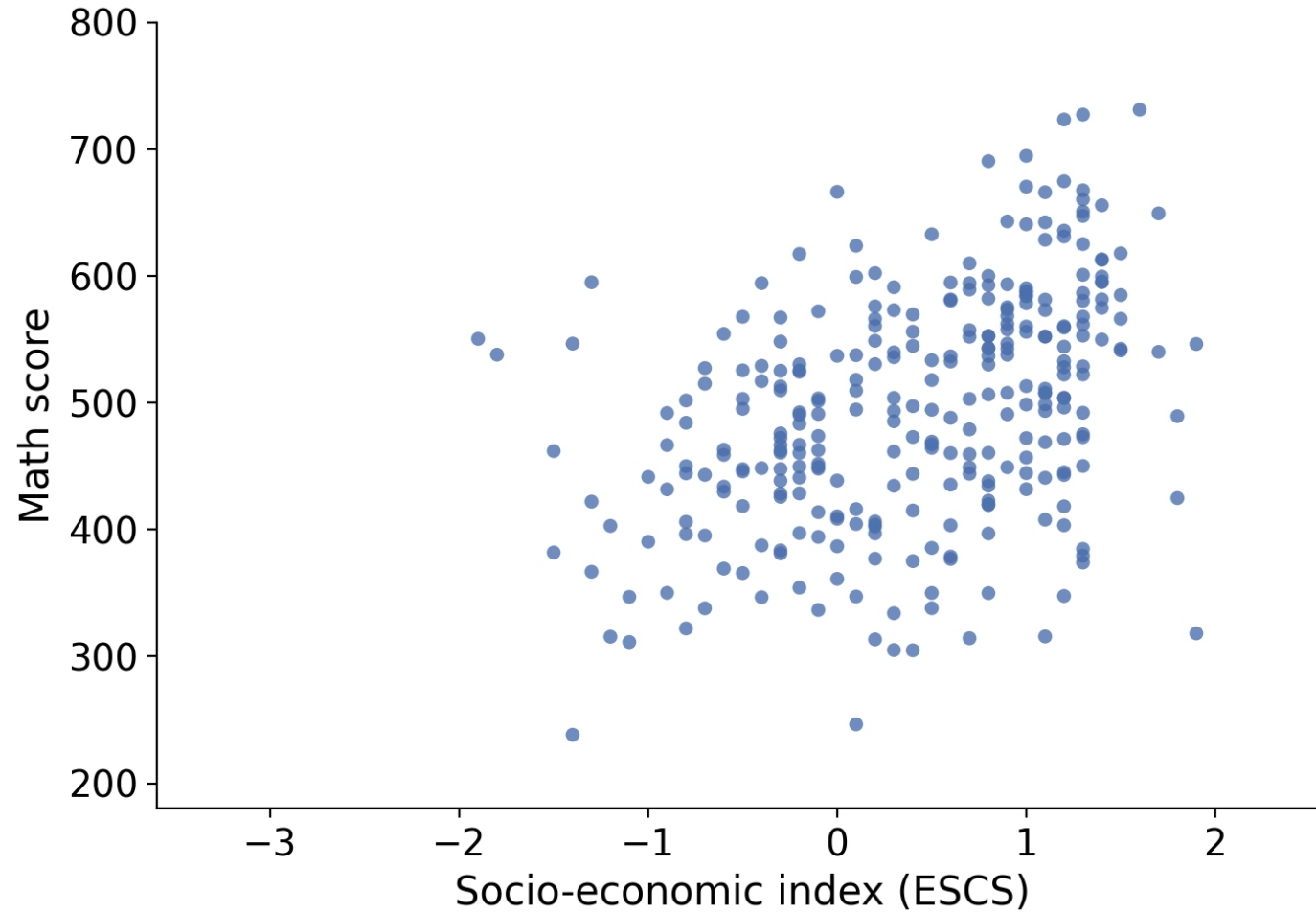


# Gallery 6: $r = -0.03$ , yet $y$ depends on $x$



Where a covariance comes from

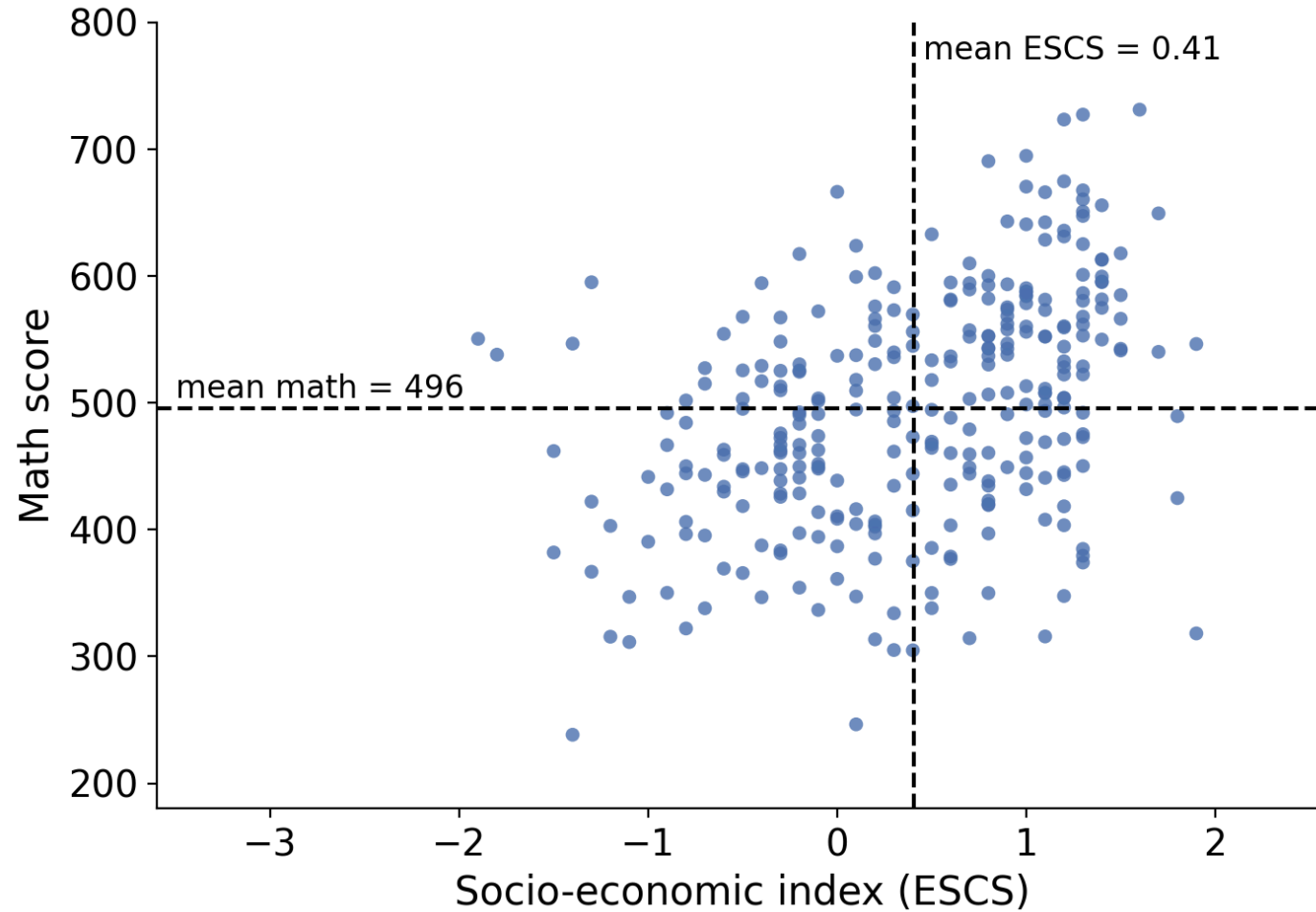
# 300 students, math against family background



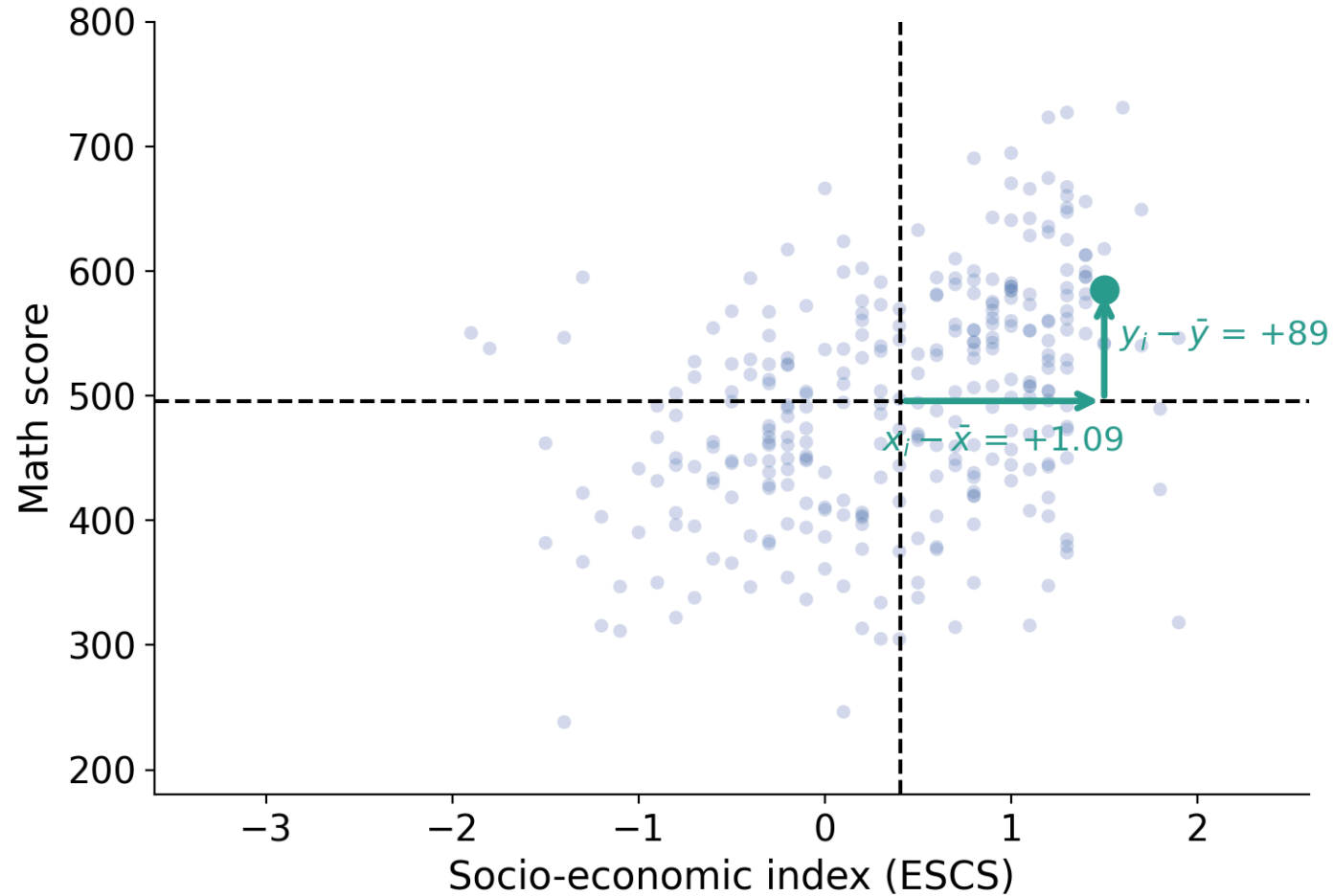
Data: PISA 2022, Australia (OECD), 300 students drawn at random



# Step 1: find the two means

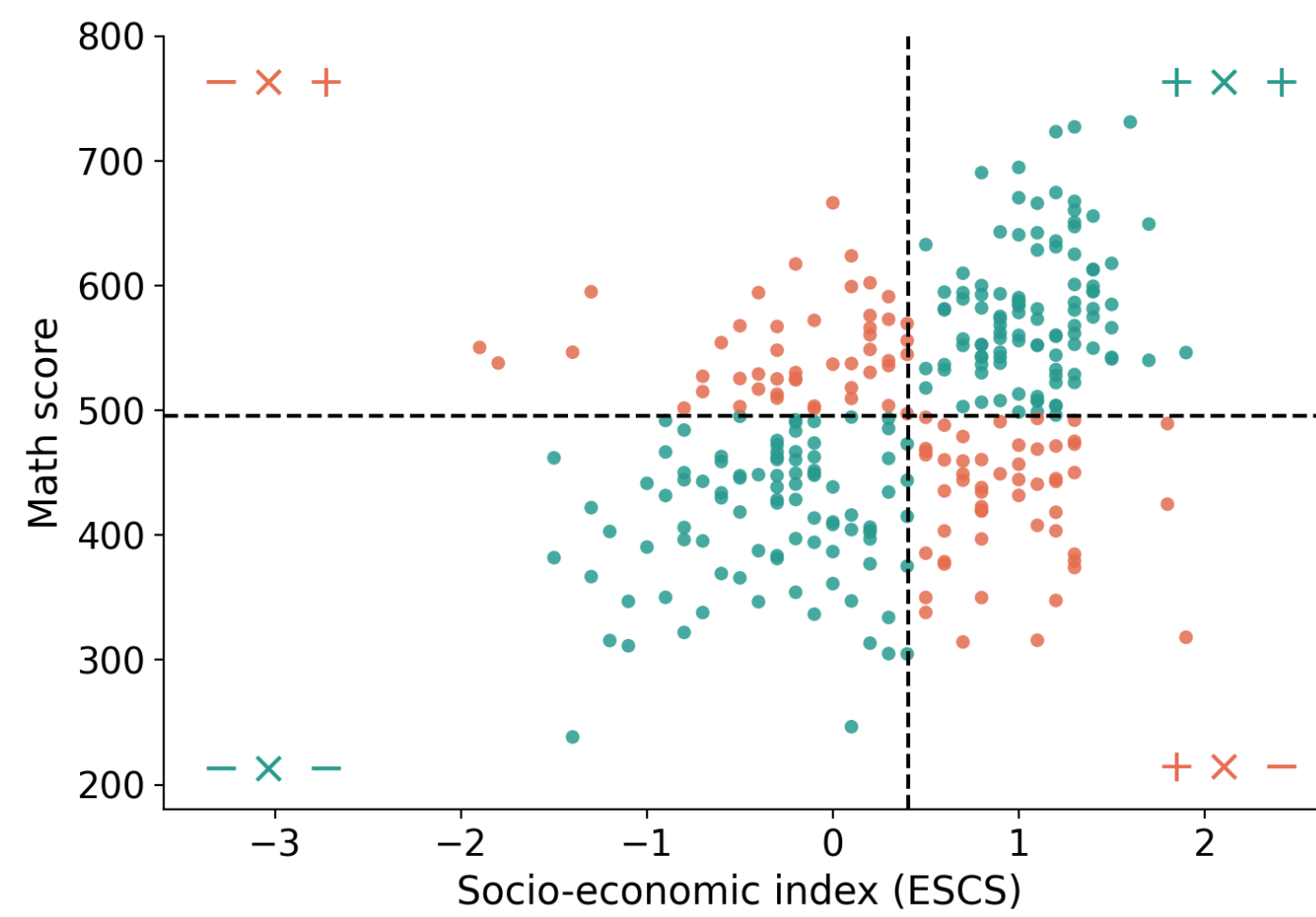


# Step 2: each student's two deviations

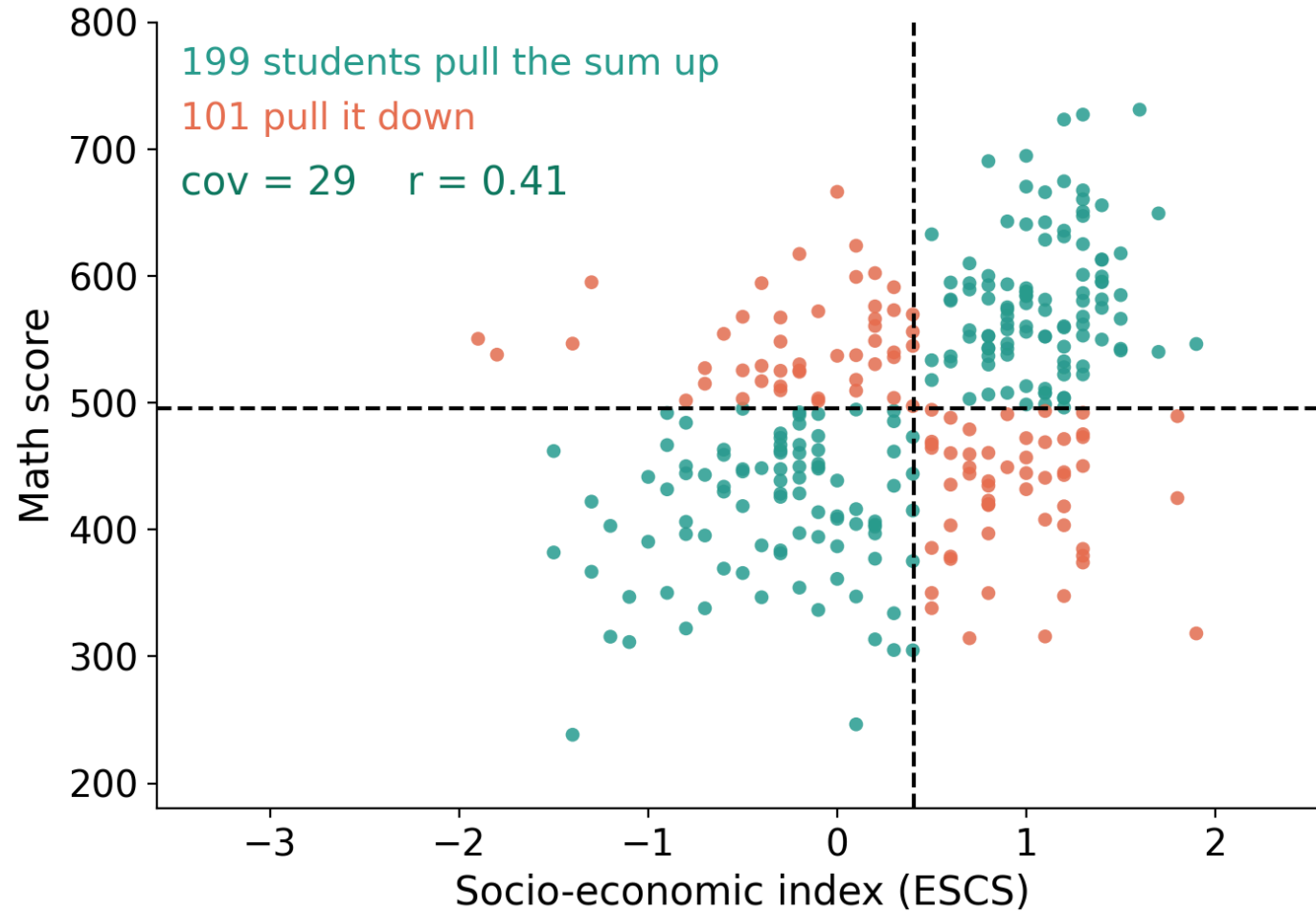


$$(x_i - \bar{x})(y_i - \bar{y}) = (+1.09)(+89) = +97$$

# Step 3: the sign of the product is the quadrant



# Step 4: add up: 199 pull up, 101 pull down



# Nonsense units, so we standardize

$$\text{cov}(X, Y) = \frac{1}{n} \sum_i (x_i - \bar{x})(y_i - \bar{y}) = 29 \text{ index points} \times \text{score points}$$

$$r = \frac{\text{cov}(X, Y)}{s_X s_Y} = \frac{29}{0.78 \times 91.7} = 0.41$$

# Some take-aways

- A conditional expectation is a function: one average for every value of  $X$
- Averaging hides the ends: 14.6 years, inside one country
- Correlation sees straight lines only, and a sample's  $r$  is itself random
- The covariance is a vote by quadrant; dividing by the SDs removes the units